Transportation Research Part C 191 (2026) 105842

Contents lists available at ScienceDirect

Transportation Research Part C

journal homepage: www.elsevier.com/locate/trc

Can coherence and coupling of quantum games get us closer to connected and autonomous lane-free traffic?☆,☆

Kinda Chakas, Lina Kattan *

University of Calgary, 2500 University Drive NW, T2Nn1N4 Calgary, Canada

ARTICLE INFO

Keywords:
Quantum-Inspired Games
Graphon Mean Field Game
Lane Free Traffic
Connected and Autonomous Driving

1. Introduction

Lane-Free Traffic (LFT) enables future Connected and Autonomous Vehicles (CAVs) to navigate seamlessly in two-dimensional spaces without the traditional lane constraints. Initially proposed by Johansen and Løvland (2015) and later formulated by Papageorgiou et al. (2021), vehicles in LFT adjust their longitudinal and lateral positions smoothly, potentially improving road efficiency. Recent studies developed new modelling and control strategies on one-directional LFT corridors (Levy & Haddad, 2022; Yanumula et al., 2023; Karalakou et al., 2023; Berahman et al., 2025). In these models, a fixed virtual boundary constrains lateral vehicle movements, ensuring vehicles stay within designated road segments while allowing lateral manoeuvres.

Despite their capacity boosting advantages, LFT presents new challenges in modelling vehicle interactions without lane discipline, as traditional car-following or lane-changing models become obsolete. LFT requires two-dimensional vehicle dynamics that accurately reflect the diverse interactions between vehicles. Therefore, new approaches are necessary to maintain LFT safety and comfort while improving its efficiency. Rostami-Shahrbabaki et al. (2023) incorporated artificial force-based models in LFT formulation to enable vehicles to "glide" safely and comfortably through decentralized collision avoidance and coordination. While force-based and potential-field approaches have been shown to be real-time feasible when interactions are restricted to a finite local neighborhood, their limitations lie primarily in their heuristic aggregation of pairwise forces and the absence of an explicit micro–macro coupling framework. Such models do not directly encode population-level equilibrium structure or density-wave propagation.

To address this, Reinforcement Learning (RL) has emerged as a promising approach for LFT trajectory management. Leurent and Mercat (2019) applied deep Q-networks to highway driving, Cao et al. (2020) developed multi-agent RL, Wang et al. (2023) implemented deep deterministic policy gradients for LFT, Zhang et al. (2024) utilized transformer-based interaction predictors, and Berahman et al. (2024) advanced multi-task RL balancing individual and collective goals. Despite their promise, RL methods still face

☆ .☆ This article is part of a special issue entitled: published in <journal name>. Please add a footnote in the below mentioned format for all special issue papers with item group description as "VSI: <special issue title>" This article belongs to the Virtual Special Issue on "special issue full title" ISTTT 2026 -TRC.

* Corresponding author.
E-mail address: lkattan@ucalgary.ca (L. Kattan).

https://doi.org/10.1016/j.trc.2026.105842
Received 1 February 2026; Received in revised form 4 May 2026; Accepted 27 June 2026
Available online 7 July 2026
0968-090X/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

ongoing challenges in ensuring safety and multi-agent convergence stability within high-dimensional, continuous action spaces. Most existing LFT frameworks primarily focus on ego vehicle navigation, overlooking the broader system-wide dynamics such as density-wave propagation and traffic stability. Micro-to-macro dynamics in lane-free traffic systems can produce system-wide properties like spontaneous synchronized flows, harmonised speeds, and optimal capacity utilisation, which ego-centric models cannot achieve. While most LFT research has focused on microscopic vehicle-level control strategies, recent studies have begun to investigate macroscopic and continuum representations of lane-free and artificial-fluid traffic, which remains an active and relatively underexplored area in LFT theory.

The challenges above motivate the search for alternative methods with stronger theoretical guarantees for large-scale interactions in LFT. Mean Field Games (MFGs), introduced by Huang et al. (2006) and Lasry and Lions (2007), offer powerful tools for addressing scalability in large systems by aggregating agent behaviours into mean fields, thereby reducing computational costs. In traffic management, MFGs have been effectively applied to vehicle routing (Huang et al., 2021), departure timing coordination (Ameli et al., 2022), and car-following trajectory optimisation (Mo et al., 2024). Notably, Mo et al. (2024) developed a second-order traffic-flow MFG framework that controls vehicle acceleration through fixed-point fictitious-play and inverse reinforcement learning, linking microscopic costs to macroscopic outcomes. However, most existing MFG models assume agent homogeneity, which fails to capture more realistic traffic heterogeneity. Graphon Mean Field Games (GMFG), introduced by Caines and Huang (2021), explicitly address heterogeneity through graph-based interaction representations. This allows GMFGs to capture complex behavioral patterns among diverse vehicles, including spatial proximity heterogeneity, directional interaction asymmetry, and cooperative strategy diversity. Preliminary lane-free GMFG applications (e.g., Chakas & Kattan, 2024) demonstrated scalability to thousands of vehicles. Yet, their approaches relied on simple interaction weights and deterministic control policies, which only indicate the magnitude of influence and limit their ability to represent CAVs' coordinated maneuvers, such as merging or lane changing. For example, when vehicles approach a merge zone, coordination demands precise timing and directional alignment, requirements that simple numerical weights cannot adequately represent.

This paper introduces a quantum-inspired GMFG approach that employs tools from quantum mechanics, but without involving actual quantum physics. Our method features three main aspects: (1) complex-valued interaction kernels that use magnitude and phase shifts to represent interaction strength and timing or directional preferences, akin to quantum wave functions; (2) adaptive coupling mechanisms that dynamically adjust overall coordination strength based on collective phase coherence, thereby enhancing cooperative behaviour when needed; and (3) interference-based coordination, where complex vehicle influences are combined (e.g., aligned phases reinforce actions) or cancelled (e.g., opposing phases negate actions), allowing for more nuanced coordination patterns. For solution algorithms, we apply actor-critic RL methods to approximate coupled Hamilton-Jacobi-Bellman (HJB) and Fokker-Planck (FP) equations. This RL approach tackles the computational challenges of solving quantum-inspired GMFG equilibria in high-dimensional LFT, enabling scalable, decentralised policy deployment while effectively managing complex-valued kernels and adaptive coupling dynamics.

This work provides significant theoretical and methodological contributions. Theoretically, this research enhances LFT theory by introducing a new quantum-inspired GMFG formulation for traffic management. The value of this Quantum approach lies in its ability to address the limitations of classical scalar models in capturing rich, phase-dependent coordination behaviours in dense, lane-free environments. By integrating complex-valued kernels and adaptive population-level coupling, we establish a novel mathematical modelling paradigm that enables interference-based coordination phenomena. Vehicles thus display interaction patterns similar to wave mechanics, leading to self-organizing merges and seamlessly smooth flow. Additional contributions include quantum-inspired mechanisms that extend the solution space beyond traditional game theory, allowing for the dynamic reinforcement or cancellation of vehicle interactions based on phase relationships. Moreover, adaptive coupling enhances cooperative driving behaviour under high phase coherence, while reverting to more selfish strategies when coordination is less effective/desirable. This creates continuous feedback loops between individual decisions and collective coordination, enabling seamless transitions during critical manoeuvres such as merging and diverging. Additionally, graphon-based modelling accurately captures spatially varied vehicle interactions across different highway zones (pre-merge, merge, cruise, diverge, and post-diverge).

Methodologically, to overcome computational time challenges and enable real-time solutions, we developed RL actor-critic neural networks for policy optimization within the complex-valued strategy space. To our knowledge, this represents the first computational method for solving quantum-inspired traffic games at this scale. This approach ensures stable optimization while enabling real-time adaptation, bridging quantum-inspired game theory and practical autonomous vehicle implementations. Notably, we prove the existence of string stability for lateral and longitudinal dimensions. The resulting framework provides the necessary robust mathematical foundation and computational tools essential for realizing cooperative CAVs driving in lane-free traffic, leading to safe, harmonious and smooth traffic.

While the proposed framework incorporates concepts from interaction modeling, cooperative behaviour, RL, and mean-field game theory, these elements are not treated as independent contributions. Instead, they form an integrated methodology centered on a single objective: decentralized control of lane-free highway traffic. Cooperative and non-cooperative interactions emerge from the GMFG formulation through local coherence mechanisms, and RL is used solely as a numerical approximation tool for solving the resulting equilibrium control problem. This unified perspective allows the problem to be addressed in a cohesive manner without introducing separate behavioural or learning models.

The rest of this paper is organized as follows. Section 2 presents a comprehensive literature review, Section 3 introduces the quantum-inspired GMFG formulation and offers a proof of the model's string stability, Section 4 reports simulation results that validate our methodology, and Section 5 concludes the study.

2. Literature review

The literature review comprises three subsections. First, we examine recent LFT research. Next, we revise GMFG-based traffic applications, focusing on approaches for modelling heterogeneous vehicular interactions on weighted networks, and discuss their initial LFT applications. Finally, we review recent developments in quantum computing related to transportation applications.

LFT has gained attention due to potential efficiency gains in road utilization and traffic management (Johansen & Løvland, 2015; Papageorgiou et al., 2021). Readers can refer to Sekeran et al. (2022) for an extensive overview of developments, policy implications, and practical considerations.

Various control strategies for CAVs in lane-free freeways have emerged, including optimal control (Yanumula et al., 2023; Levy & Haddad, 2022), potential line policies (Zhang et al., 2023), and ad-hoc vehicle kinematics (Malekzadeh et al., 2022). Decentralized nonlinear feedback control has also been proposed for traffic systems inspired by artificial-fluid and continuum traffic-flow models. In particular, Karafyllis et al. (2018) developed feedback-based stabilization laws for nonlinear traffic dynamics, providing rigorous guarantees on stability and convergence using locally available state or boundary information. Relatedly, distributed model predictive control (MPC) has been explored for coordinating automated vehicles in lane-free traffic. Dabestani et al. (2025) showed that distributed MPC with limited prediction horizons and neighborhood-based coupling enables real-time feasible coordination while explicitly respecting safety and actuation constraints, offering a principled alternative to purely reactive control laws. Recent studies also explored vehicle flocking (Rostami-Shahrbabaki et al., 2023) and integration of human-driven vehicles (Syed et al., 2025). LFT concepts are now transferred into arterial applications, including intersection design (Naderi et al., 2024) and interaction with vulnerable road users (Malcolm & Bogenberger, 2025).

While most LFT research has focused on microscopic vehicle-level control strategies, recent studies have begun to address macroscopic and continuum representations of lane-free and artificial-fluid traffic, including macroscopic flow calibration (Papamichail et al., 2023), continuum artificial-fluid formulations linking congestion and control (Karafyllis et al., 2025), and numerical studies of macroscopic automated-vehicle traffic models (Titakis et al., 2024). However, these approaches primarily model aggregate traffic dynamics and do not embed microscopic vehicle decision-making within a unified equilibrium-based micro–macro feedback framework. This gap motivates equilibrium-based formulations, such as mean-field and graphon mean-field games, which explicitly couple individual vehicle optimization with macroscopic traffic density evolution.

Reinforcement learning based methods are mainly used in lane-free environments and have shown promise in reducing collisions, increasing efficiency, and improving coordination. Berahman et al. (2025) combined cruise control with deep deterministic policy gradients (DDPG) for proactive collision avoidance. Karalakou et al. (2023) developed coordinated lateral and longitudinal decision-making using DDPG, focusing on multi-dimensional control. Troullinos et al. (2021) employed multi-agent DRL with dynamic graph structures to improve inter-vehicle coordination on freeways. They also explored reward-function design within a Markov decision process framework, comparing Deep Q-Network and DDPG performances to highlight the critical role of reward architecture in determining agent effectiveness and robustness, while also noting the limitations of rule-based heuristics.

Decentralized nonlinear feedback control has also been proposed for traffic systems inspired by artificial-fluid and continuum traffic-flow models. In particular, Karafyllis et al. (2018) developed feedback-based stabilization laws for nonlinear traffic dynamics, providing rigorous guarantees on stability and convergence using locally available state or boundary information. Relatedly, distributed model predictive control (MPC) has been explored for coordinating automated vehicles in lane-free traffic. Dabestani et al. (2025) showed that distributed MPC with limited prediction horizons and neighborhood-based coupling enables real-time feasible coordination while explicitly respecting safety and actuation constraints, offering a principled alternative to purely reactive control laws.

Moreover, these studies mainly depend on scalar interaction models or discrete reward functions, which do not fully capture continuous-phase vehicle coordination in lane-free scenarios. Scalar models simplify inter-vehicle influences into numerical weights, often missing the timing, sequencing, and phase alignment of maneuvers. Additionally, discrete reward frameworks divide behaviours into separate events, which is not ideal for modelling the detailed temporal coherence essential for situations like zipper merges or coordinated platoon formations. As a result, these methods face challenges with accurate synchronization and phase-dependent interactions that are vital for overall traffic efficiency. Here comes the role of GMFG in overcoming the above limitations and capturing continuous, heterogeneous interactions.

GMFG generalizes classical mean field games by replacing the homogeneous “average agent” with a graphon, a measurable function representing heterogeneous pairwise influences in large populations. Introduced by Caines and Huang (2021), GMFG preserves analytical tractability while explicitly encoding variations in interaction strength, directionality, and agent type, making them effective for modelling non-uniform and non-local interactions.

Subsequent theoretical work established well-posedness, ε-Nash convergence, stability properties, and scalable numerical solvers (Gao & Caines, 2019), showing that graphons effectively model non-local contagion, asymmetric influences, and mixed state spaces. Transportation researchers recently applied GMFG to lane-free freeway traffic; Chakas and Kattan (2024) modelled heterogeneous vehicle classes and anisotropic interactions, demonstrating scalability to thousands of CAVs while preserving solution fidelity.

Making graphon-based approaches applicable in real-time control demands new computational methods, what emerging quantum algorithms provide. Quantum computing algorithms deliver computational advantages to addressing transportation challenges. Recently, quantum computing applications in transportation include congestion forecasting using Temporal-Spatial Quantum Graph Convolutional Networks (Qu et al., 2022) and Quantum Neural Networks (Moskvin, 2023), hyperparameter optimization via Quantum Particle Swarm Optimization (Zhang et al., 2021), and adaptive signal timing using Quantum Annealing with QUBO formulations (Singh et al., 2021; Inoue et al., 2021; Marchesin et al., 2023). Quantum Genetic Algorithms and Quantum-Inspired Firefly Algorithms

were also shown to enhance real-time traffic management by optimizing network signals dynamically (Acampora et al., 2023; Upare et al., 2024). Despite hardware limitations, vehicle routing and logistics problems benefit from quantum variational algorithms, annealing, and hybrid approaches, efficiently tackling complex optimization problems (Bentley et al., 2022; Azad et al., 2022; Harikrishnakumar et al., 2022; Phillipson, 2024; Fitzek et al., 2024). In addition, quantum-enhanced machine learning, such as Quantum LSTM networks, supports autonomous vehicle perception and platooning, alongside quantum-secure cybersecurity solutions (Harwood et al., 2021; Luckow et al., 2021).

To the authors' knowledge, these are the only quantum computing applications in transportation domain to date. Notably, key quantum concepts, such as quantum game frameworks with coherence-based coupling, and superposition-driven strategies, have not yet been applied to large-scale transportation problems formulated within a theoretical game framework. While quantum-inspired methods have advanced classical game theory by enabling correlated equilibria and resolving classical dilemmas (Eisert et al., 1999; Piotrowski & Sładkowski, 2003; Kolokoltsov, 2022), their potential in transportation games remain largely uncharted.

3. Methodology

MFG theory provides a mathematical framework for modelling strategic interactions in large populations of agents, where each agent's optimal behaviour depends on aggregate population states rather than on pairwise interactions (Lasry & Lions, 2007; Huang, Caines, & Malhamé, 2006). In a standard MFG, equilibrium is defined through the coupling of two equations: (i) an HJB equation that characterizes the optimal control policy of a representative agent, and (ii) a FP or continuity equation describing the evolution of the population density under the optimal policy. An MFG equilibrium is achieved when the population density used in the agent's optimization matches the density generated by the resulting optimal policies, forming a fixed point between microscopic decision-making and macroscopic population evolution.

GMFGs extend classical MFGs by allowing interaction strengths to vary across a continuum of agent types or network positions, represented by a graphon function (Caines & Huang, 2021; Parise and Ozdaglar, 2023). Rather than assuming symmetric or mean-homogeneous interactions, GMFGs capture structured and heterogeneous coupling, such as those arising from spatial geometry, behavioural phases, or road-segment topology. In this setting, the mean-field term is no longer a simple average over the population but an integral weighted by the interaction kernel, enabling richer models of coordinated behaviour and nonlocal influence.

The present work builds on these ideas by formulating a lane-free traffic GMFG in which individual vehicles solve an HJB problem influenced by macroscopic traffic density through: (1) a complex-valued interaction amplitude encoding segment structure, destination alignment, and behavioural coherence; and (2) an adaptive coherence mechanism that reinforces alignment with dominant traffic streams. The FP equation governing density evolution incorporates these microscopic control responses, producing a closed-loop micro–macro feedback system.

The main objective of our approach is to generate continuous CAV trajectories (longitudinal acceleration and steering angle) that ensure system-wide safety and efficiency. Fig. 1 illustrates the system architecture, distinguishing between the offline learning phase and the online operational phase.

- Offline Phase (Centralized Learning & Calibration):
  - Scope: Centralized.

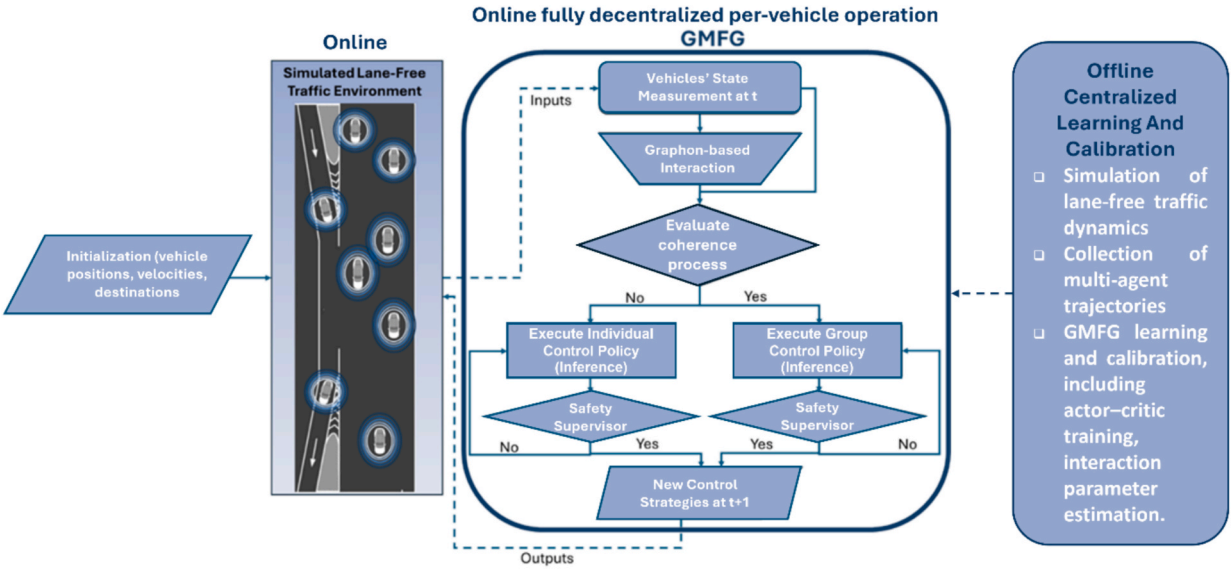


Fig. 1. Flowchart of the system architecture.

- Process: A high-fidelity lane-free traffic simulation is used to model global dynamics and generate multi-agent trajectories. In this phase, the Graphon Mean Field Game (GMFG) solution is approximated using Actor-Critic Reinforcement Learning. The policy parameters are iteratively optimized using data collected from the simulation until the actor-critic networks converge to the game equilibrium.
- Output: This phase produces a fixed, calibrated control policy (Actor Network) and interaction kernel parameters.

- Online Phase (Decentralized Execution):
  - Scope: Decentralized Onboard Execution.
  - Process: Upon deployment, traffic operation is fully decentralized. Each vehicle operates independently using the pre-trained policy embedded in its onboard computer. At each time step, the operation cycle is as follows:
    1. State Measurement: The vehicle captures its local state (position, velocity) and estimates the state of neighboring vehicles, defined as those vehicles with non-negligible interaction weights under the spatially decaying kernel.
    2. Graphon Field Estimation: The vehicle computes the local interaction term using the pre-calibrated kernel parameters derived offline.
    3. Policy Execution (Inference): Instead of running an iterative optimization loop, the vehicle performs a rapid forward pass through its pre-trained Actor Network. This efficiently maps the high-dimensional aggregated state directly to optimal control inputs.
    4. Safety Supervisor: The proposed control actions pass through a safety layer to guarantee collision avoidance before being sent to the actuators.
  - Computational Feasibility: This inference-based approach incurs negligible computational cost, ensuring real-time responsiveness without the need for centralized coordination or live training.

Details of these steps and their mathematical formulation are provided in Section 3.1. Section 3.2 presents mathematical proof of traffic stability in both the lateral and longitudinal dimensions.

Details of these steps and their mathematical formulation are provided in Section 3.1. Section 3.2 presents mathematical proof of traffic stability in both the lateral and longitudinal dimensions.

3.1. *Quantum-inspired GMFG formulation*

Section 3.1 outlines the GMFG framework on an LFT freeway segment with two on-ramps and two off-ramps. The GMFG formulation integrates multiple critical elements structured as a tuple comprising: (1) Spatial Domain, State Space, and Action Space, (2) Vehicle Dynamics, (3) Interaction Kernel, (4) Cost Function, (5) Adaptive Coherence Mechanism, (6) GMFG Equilibrium Conditions, and (7) Numerical Solution Algorithm: Reinforcement-Learning–Based Solver.

3.1.1. *Spatial domain, state space, and action space*

The spatial domain defines the freeway segment with specified length and width, with vehicles interacting at designated segments as shown in Fig. 2. The action space encompasses out the feasible bounds of vehicle acceleration.

Let the spatial domain be defined by Equation (1):

$$D = \{(x,y)\,|0 \le x \le L, 0 \le y \le W\}, \tag{1}$$

Which describes the freeway of length L and width W. The corridor width W is treated as the fixed effective lateral operating domain of the mainline lane-free roadway. Ramp-related merging and diverging effects are represented through region labels, transition rules, and local interaction kernels rather than by changing the lateral boundary condition. Therefore, all vehicle lateral positions remain constrained within $0 \le \mathrm{y} \le \mathrm{W}$ throughout the simulation. Road boundaries are enforced as hard constraints by restricting the admissible lateral state and control space such that each vehicle's lateral position $y_i(t)$ satisfies $0 \le y_i(t) \le W$ for all t, and any control input that would violate these bounds is either rendered infeasible or projected onto the admissible set during both offline training and online execution. On-ramps are located at longitudinal coordinates $x_{on,1}$, $x_{on,\ 2}$ and off-ramps at $x_{off,1}$, $x_{off,\ 2}$. Vehicles may enter from: upstream mainline at (labelled origin $O_i = 0$), on–ramp one at (labelled $O_i = 1$), or on-ramp two at (labelled $O_i = 2$). Similarly, exits occur at: downstream mainline at (labelled destination $D_i = 0$ off–ramp 1 at (labelled $D_i = 1$), or off–ramp two at (labelled $D_i = 2$).

The freeway is partitioned into these five segments to capture distinct driving regimes, pre-merge, merging, cruising, diverging, and

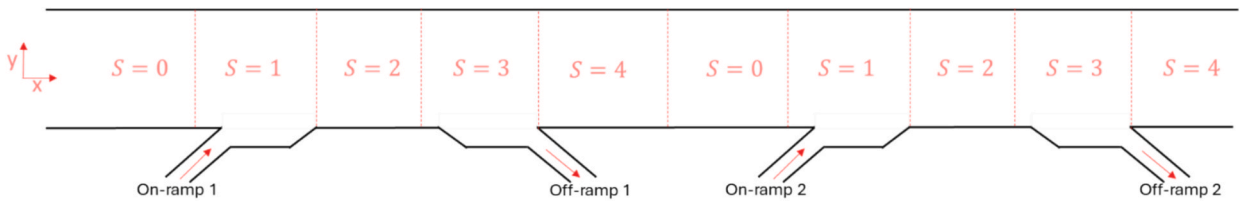


Fig. 2. Example of a Longitudinal Segmentation of Freeway Regions.

post-diverge, each with its own interaction laws and region-transition rules that automatically switch the vehicle's control logic as it crosses from one segment into the next. Discrete segments indices along the longitudinal axis are defined by Equation (2) and illustrated in Fig. 2:

$$S = \left\{ \begin{array}{c} 0, pre-mergemainline \\ 1, mergingzone \\ 2, cruisingzone \\ 3, divergingzone \\ 4, post-divergemainline \end{array} \right\} \tag{2}$$

The above segmentation captures the physical structure of a corridor with a merging on-ramp and a diverging off-ramp, and it enables phase-specific collision-avoidance and merge/diverge driving rules while preserving Lipschitz continuity of the transition map across region boundaries. Each of the five regimes corresponds to a distinct driving protocol that reflects the underlying roadway geometry. In the pre-merge region, vehicles maintain larger following gaps and rely primarily on longitudinal speed coordination to stabilize flow. Upon entering the merge buffer, typically a fixed distance upstream of the junction, such as the 200-m stretch between the on-ramp warning sign and the physical ramp entrance, the vehicle automatically activates a merge controller that reduces headways when necessary in dense traffic, increases lateral responsiveness, and engages a predictive gap-selection routine to accommodate incoming on-ramp vehicles. After the merge point, vehicles transition into a cruising regime where lateral motion is minimized, and longitudinal consensus control promotes maximum throughput. As they approach their designated exit, the diverge controller gradually restores lateral freedom, reduces speed to align with the off-ramp geometry, and applies anticipatory steering to guide vehicles into the diverge lane. Finally, in the post-diverge region, vehicles return to moderate headways and a primarily longitudinal coordination mode. Crucially, these behaviours do not represent independent control subsystems. The region index $s_i \in \{0, \cdots, 4\}$ is embedded directly in the agent's state vector $\chi_i(t)$ and evolves deterministically, meaning all vehicles remain part of a single coupled system governed by one GMFG controller. Segmentation therefore functions as a mode identifier that determines which driving protocol is active at a given physical location, allowing vehicles to adjust behaviour seamlessly without abrupt manoeuvres, while ensuring that the overall dynamics remain continuous, smoothly bounded, and analytically well-posed across the entire corridor.

The agent state of the vehicle i is given by Equation (3):

$$\chi_i(t) = (x_i(t), y_i(t), v_i(t), \psi_i(t), S_i, O_i, D_i) \in \chi, \tag{3}$$

where (x_i, y_i) are the planar coordinates in the corridor-aligned global frame, v_i is the scalar speed, ψ_i is the heading angle, and S_i, O_i, D_i are discrete parameters.

The continuous components are bounded as $x \in [0, L], y \in [0, W], v \in [0, v_{max}], \psi \in [-\psi_{max}, \psi_{max}]$.

Equipped with the product topology, χ is compact. This follows from Tychonoff's theorem: a finite Cartesian product of compact metric spaces is compact (Munkres, 2000). Compactness ensures the tightness of probability measures on χ and allows the GMFG fixed-point arguments to be developed in a well-posed topological setting.

Each CAV controls its longitudinal acceleration and steering angle:

$$A_i(t) = (a_i(t), \delta_i(t)) \in \mathscr{A} = [a_{min}, a_{max}] \times [\delta_{min}, \delta_{max}] \tag{4}$$

where δ_i is the front-wheel steering angle. Equation (4) is subject to the jerk bound $|a_i(t + \Delta t) - a_i(t)| \leq j_{max} \Delta t$. Thus, $\mathscr{A}$ is a compact, convex subset of $\mathbb{R}^2$.

3.1.2. Vehicle dynamics

To capture physically consistent two-dimensional motion, we model each CAV using the kinematic bicycle model (Rajamani, *Vehicle Dynamics and Control*, 2012).

This model couples longitudinal and lateral motion through the heading dynamics, preventing any artificial decoupling of the x- and y-directions.

The continuous-time dynamics are: $\dot{x}_i = v_i \cos\psi_i, \dot{y}_i = v_i \sin\psi_i, \dot{v}_i = a_i, \dot{\psi}_i = \frac{v_i}{\ell} \tan\delta_i$,

where ℓ is the wheelbase.

These equations describe car-like planar motion and are Lipschitz on the compact state–action domain, satisfying the standard McKean–Vlasov regularity assumptions required for GMFG existence and uniqueness (Carmona & Delarue, 2018).

For the GMFG formulation, we employ a fixed-step Euler discretization, where for each agent i:

$$x_i(t + \Delta t) = x_i(t) + v_i(t).cos\psi_i(t)\, \Delta t, \tag{5}$$

$$y_i(t + \Delta t) = y_i(t) + v_i(t).sin\psi_i(t)\, \Delta t,$$

$$v_i(t + \Delta t) = v_i(t) + a_i(t)\, \Delta t,$$

$$\psi_i(t + \Delta t) = \psi_i(t) + \frac{v_i(t)}{\ell} tan\delta_i(t)\, \Delta t.$$

The discrete region's index updates via:

$$S_i(t+1) = \Gamma(S_i(t), x_i(t+1)), \tag{6}$$

Where Γ is a piecewise thresholding function that updates the segment label when the vehicle crosses predefined longitudinal boundaries at points $0, x_{on,1}, x_{on,2}, x_{off,1}, x_{off,2}, L$.

Denoting the map $F : \chi \times \mathscr{A} \to \chi$, one checks Lipschitz continuity in both arguments, ensuring well–posedness of the McKean–Vlasov limit (Carmona & Delarue, 2018). This deterministic evolution of the region index ensures that transitions between neighboring regions occur smoothly as vehicles move longitudinally along the corridor. Because s_i changes only when the vehicle crosses predetermined spatial thresholds, and because the mapping is Lipschitz, no discontinuities are introduced in the state evolution. Thus, segmentation enriches the spatial structure of the corridor while preserving the regularity of the overall dynamics.

It is important to emphasize that heading is a state variable dependent on the integration of the steering input. This formulation ensures that all generated trajectories are dynamically feasible by construction, as the discrete transition $\chi_i(t)$ to $\chi_i(t+\Delta t)$ (Eq. (5) strictly obeys the non-holonomic constraints of the bicycle model.

3.1.3. Interaction kernel

Within the GMFG framework, the interaction kernel defines how an individual vehicle's dynamics are affected by the aggregate behaviour of the neighboring traffic population. Quantum kernels and amplitude-based similarity measures have deep roots in quantum information theory. Wootters (1981) first showed that statistical distances can be represented through Hilbert-space amplitudes, highlighting how phase and magnitude jointly encode relational structure. This idea later supported quantum associative memories (Ventura & Martinez, 2000), where complex amplitudes capture similarity through interference, and ultimately motivated modern quantum kernel methods for large-scale learning (Rebentrost et al., 2014). Inspired by this lineage, our formulation transfers these amplitude-based principles to a lane-free GMFG setting: the magnitude of the kernel encodes spatial and contextual proximity, while the phase captures behavioural coherence between vehicles (e.g., heading alignment, destination compatibility, and segment progression). This allows the interaction weight to reflect not only how strongly two vehicles should influence each other, but also how their intended motions align or conflict, enabling constructive or destructive interference in a manner analogous to quantum kernels yet tailored to traffic dynamics.

Concretely, in a lane-free corridor, each CAV modulates its longitudinal acceleration and steering command based on a weighted aggregate of neighbors' states, where the kernel prescribes the strength and form of those weights. Accordingly, the interaction kernel encodes heterogeneous, distance-weighted, and directionally anisotropic influences among vehicles; consequently, agents within the same spatial neighborhood do not exert equal influence, as coupling strength depends on relative distance, heading alignment, destination compatibility, region index, and phase coherence rather than on uniform proximity or explicit communication reliability. To reflect realistic driving, our kernel combines four principal effects:

- an exponential spatial decay that makes distant vehicles nearly invisible (i.e. not influencing the behaviour of the ego vehicle),
- a directional bias that favors the impact of vehicles ahead,
- a smoothly varying region-coupling term that activates distinct merge, cruise, and diverge controllers in the five road phases described in section 3.1.1, and
- a destination-coherence factor that strengthens grouping among vehicles sharing the same exit.

Thus, each vehicle computes a mean-field amplitude by integrating this complex valued kernel over the current state distribution. This amplitude captures the collective influence of surrounding traffic and is used as an input to the vehicle optimal control law.

We now introduce the following components of the interaction amplitude, $\widetilde{K}\left(\chi_i, \chi_j\right)$:

1- The first component of our interaction kernel is an exponential spatial decay represented by Equation (7):

$$exp\left(-\alpha . r_{i,j}^{p}\right), \alpha > 0, p \geq 1, \tag{7}$$

Where,

$$r_{i,j} = \sqrt{\left(x_j - x_i\right)^2 + \left(y_j - y_i\right)^2},$$

is the Euclidean distance between vehicle i and vehicle j. The parameter α sets the characteristic interaction range, larger α causes influence to drop off more rapidly, while the exponent p adjusts the sharpness of that decay.

To encode directionality in a smooth manner, each neighbor is weighted by a bearing function that emphasizes forward interactions:

$$D\left(\theta_{ij}\right) = 0.65 + 0.35cos\left(\theta_{ij}\right), with \theta_{ij} = arctan\left(\frac{y_j - y_i}{x_j - x_i}\right) \tag{8}$$

The constants in Equation (8) were chosen through thorough sensitivity analysis to yield a smooth, strictly positive, and bounded directional weighting that emphasizes forward interactions while avoiding degenerate coupling. This equation yields $D = 1.0$ when vehicle j lies directly ahead ($\theta_{ij} = 0$), and $D = 0.3$ when it is directly behind ($\theta_{ij} = \pm\pi$), thereby preserving a nonzero rear-view influence. For side-by-side alignments ($\theta_{ij} = \pm\frac{\pi}{2}$), the weight smoothly interpolates to 0.65. The resulting sinusoidal profile models

realistic sensor fields of view, giving the strongest influence on vehicles in front while still accounting for those at lateral or rear positions without introducing discontinuities.

2- Second, segment coupling is encoded by:

$$R(\Delta S) = exp\big(-\gamma\big|S_i - S_j\big|\big), \gamma > 0, \tag{9}$$

The parameter γ is the segment-coupling decay rate that determines how quickly the interaction strength drops off as the segment difference increases. The equation smoothly attenuates the interaction strength across different traffic segments, from full coupling ($\Delta S = 0$) to exponentially reduced coupling as ΔS increases.

3- Third, destination-coherence weighting enhances the interaction of vehicles sharing the same exit target:

$$E\big(D_i, D_j\big) = 1 + \kappa_0 1_{\{D_i = D_j\}}, \kappa_0 \geq 0. \tag{10}$$

This factor raises the kernel by κ_0 whenever two vehicles in the same segment plan to exit at the same location.

4- Fourth, near merging or diverging boundaries a lateral nudging field guides vehicle trajectory. In this paper, "nudging" refers to a smooth, distance-based lateral guidance force that gently steers each vehicle toward the correct merge or diverge segment (and away from incorrect ones) before it reaches the ramp.

Let $d_{on,k}(x)$ denotes the distance from a point x along the roadway centerline to the k-th on-ramp merge boundary and let $d_{off,l}(x)$ denotes the distance to the l-th off-ramp diverging boundary. Specifically, if $x_{on,k}^{start}$ and $x_{on,k}^{end}$ mark respectively the longitudinal entrance and exit of the merge zone for the on-ramp k, then for any vehicle at longitudinal coordinate x:

$$d_{on,k}(x) = \begin{cases} x - x_{on,k}^{start}, x_{on,k}^{start} \leq x \leq x_{on,k}^{end} \\ x - x_{on,k}^{start}, x < x_{on,k}^{start} \\ x_{on,k}^{end} - x, x > x_{on,k}^{end} \end{cases} \tag{11}$$

Thus $d_{\mathrm{on},k}(x) > 0$ whenever the vehicle lies inside the merge region, and $d_{\mathrm{on},k}(x) < 0$ otherwise. An analogous definition holds for $d_{off,l}(x)$ with the diverge boundaries $x_{off,l}^{start}$ and $x_{off,l}^{end}$.

The lateral nudging term $N\big(X_i, X_j\big)$ between two vehicles i and j functions as a continuous lateral guidance field that encourages vehicles to align smoothly within merging or diverging corridors. Vehicles inside the merge region exert a gentle lateral attraction on approaching vehicles, guiding them into available gaps, while those outside the region apply a repelling force to discourage unsafe entries. This push–pull dynamic mimics mathematically human driving behaviour. At the microscopic level, nudging reduces sharp lateral accelerations by allowing each vehicle to anticipate the movement of surrounding vehicles within a graded influence field. This results in smoother steering, lower accelerations, and fewer abrupt maneuvers. Collectively, these effects promote self-organized streams formation and coordinated gap usage near merges. Macroscopically, the nudging field increases throughput near merging segments and suppresses shockwaves by enabling cooperative alignment over a spatial buffer zone defined by decay rates η and $\eta_\perp$. In incorporating this decay term, each vehicle is subjected to a smoothly diminishing influence from others as distance in x or y grows, ensuring that nearby vehicles contribute more to its steering decision. This spatial distribution is later shown to reduce downstream congestion and stabilizes density profiles. From a control-theoretic view, nudging enhances lateral string stability by creating a distributed feedback mechanism. Vehicles adjust their lateral positions based on a weighted average of nearby agents, preventing small disturbances from growing and maintaining stability in two-dimensional traffic flow.

The nudging influence over the ramp location is:

$$N(X_i, X_j) = \begin{cases} exp\left(-\frac{\big|d_{on,k}(x_i)\big|}{\eta}\right).exp\left(-\frac{\big|y_j - y_i\big|}{\eta_\perp}\right).sign\big(d_{on,k}(x_j)\big), k = 1, 2 for on - ramp \\ exp\left(-\frac{\big|d_{off,l}(x_i)\big|}{\eta}\right).exp\left(-\frac{\big|y_j - y_i\big|}{\eta_\perp}\right).sign\big(d_{off,l}(x_j)\big), l = 1, 2 for off - ramp \end{cases} \tag{12}$$

here, $\exp\left(-\frac{d_{on,k}(x_i)}{\eta}\right)$ decays the influence with the subject vehicle's longitudinal distance from the merge zone. The parameter $\eta > 0$ sets the rate this longitudinal influence fades away. $\exp\left(-\frac{|y_j - y_i|}{\eta_\perp}\right)$ attenuates the influence with lateral separation between the two vehicles. The parameter $\eta_\perp > 0$ controls the lateral decay rate. $sign\big(d_{on,k}(x_i)\big)$ determines the direction of the nudge. If vehicle j lies *inside* the merge zone for the ramp k, $d_{on,k}(x_i) > 0$, then $sign = +1$, so j nudges i *toward* the merge corridor. If j lies *outside* $d_{on,k}\big(x_j\big) < 0$, then $sign = -1$, so j pushes i *away* from the merge. The off-ramp sum is identical in form, using $d_{off,l}$ in place of $d_{on,k}$.

To summarize, the classical coupling kernel is thus expressed as K_{cl}:

$$K_{cl}\big(\chi_i, \chi_j\big) = D\big(\theta_{ij}\big).exp\Big(-\alpha.r_{i,j}^p\Big).exp\big(-\gamma\big|S_i - S_j\big|\big).\Big[1 + \kappa_0 1_{\{D_i = D_j\}}\Big] + \lambda N\big(X_i, X_j\big) \tag{13}$$

λ weights the overall strength of the nudging influence.

In Equation (13), the classical coupling kernel $K_{cl}\left(\chi_i,\chi_j\right)$ succinctly encodes the multidimensional influences exerted by vehicle j on vehicle i into a single scalar weight. The kernel's form ensures that: (1) angular weighting emphasizes forward-looking interactions while preserving nonzero rear and lateral contributions; (2) an exponential function of Euclidean separation provides a smooth, tunable cut-off for long-range effects; (3) segment-based attenuation modulates intervehicle coupling according to their relative positions within merge, cruise, or diverge segments; (4) a binary, destination-based multiplier enhances affinity among vehicles sharing a common downstream ramp; and (5) an additive, distance- and sign-sensitive lateral "nudging" term furnishes anticipatory steering guidance near on– and off-ramps. When integrated against the instantaneous state distribution, K_{cl} yields a continuous mean-field amplitude that effectively represents the aggregate "traffic pressure" on the ego vehicle. This amplitude is directly ingested by the vehicle's optimal control law, enabling smooth, cooperative adjustments in longitudinal acceleration and steering (and thus lateral motion) without abrupt discontinuities.

3.2. Quantum-inspired interaction kernel: amplitude, phase, superposition, and interference

To extend classical mean-field interactions with quantum-inspired behaviour, the classical coupling kernel (equation (13) is multiplied by a unit-modulus phase factor, AKA as complex amplitude, $exp\left(i\beta\phi\left(X_i,X_j\right)\right)$, where $\phi\left(X_i,X_j\right)$ constitutes a bounded, Lipschitz phase function. This transformation represents a paradigmatic shift from scalar interaction weights to complex probability amplitudes, fundamentally altering how vehicle influences propagate through the traffic. The unit modulus property $\left|exp\left(i\beta\phi\left(X_i,X_j\right)\right)\right|=1$ means this factor rotates the complex weight without changing its magnitude, ensuring that the classical amplitude K_{cl} remains bounded and unaltered. In practice, the unit–modulus factor serves to encode alignment or opposition information by rotating each neighbor's influence vector in the complex plane, allowing cooperative signals to reinforce each other and conflicting signals to partially cancel, without affecting the underlying interaction strength. The smooth dependence on ϕ ensures Lipschitz continuity in X_i, X_j, a critical property for proving existence and uniqueness of the mean–field equilibrium, while the exponential map linearizes phase addition into multiplication, facilitating both theoretical analysis and numerical implementation without altering the kernel's magnitude.

In quantum mechanics, an amplitude is a complex number whose modulus squared gives a probability density. The resulting quantum-inspired kernel, $\widetilde{K}\left(\chi_i,\chi_j\right)$, is a probability-amplitude for how strongly vehicle j influences vehicle i. While the modulus $|\widetilde{K}|=K_{cl}$ recovers the classical interaction strength, its argument (phase) incorporates additional qualitative and behavioural information (e.g. interference of alignment information), which cannot be captured by a real weight alone. In other words, the classical GMFG captures only the magnitude of interactions between vehicles, treating all influences of equal strength identically regardless of their directional compatibility or temporal coordination. This scalar formulation is inadequate for modeling modern CAV systems where vehicles possess sophisticated V2V communication capabilities, enabling them to share trip purposes, destinations, merge intentions, and real-time coordination strategies. The quantum-inspired extension transcends this limitation by encoding cooperative versus competitive relationships through phase relationships, enabling the emergence of self-organizing, coherent traffic patterns such as coordinated zipper merges and adaptive stream flow formations, while preserving the analytical tractability necessary for GMFG existence and uniqueness proofs.

Hence, for any pair of CAVs i and j, we define a phase function: $\phi\left(X_i,X_j\right):\chi\times\chi\rightarrow[-\pi,\pi]$, normalized so that $\phi=0$ when the two vehicles exhibit identical merge progress or heading alignment (perfect synchrony), and $\phi=\pm\pi$ when they operate in maximal opposition (exemplified by one vehicle initiating merge entry while another completes merge exit).

A representative phase function incorporating both spatial and directional information is:

$$\phi\left(X_i,X_j\right)=\omega_\psi\left(\psi_j-\psi_i\right)+\omega_D\,1_{\left\{D_i\neq D_j\right\}}+\omega_b arctan\left(\frac{y_j-y_i}{x_j-x_i}\right)-\psi i, \tag{14}$$

where $\omega_\psi,\omega_D,\omega_b\in\mathbb{R}$ are tuning parameters that weight heading misalignment, destination mismatch, and geometric bearing, respectively. This phase design ensures that the dominant contributions reflect behavioural coherence, while geometric location plays a secondary but informative role. The term $\psi_j-\psi_i$ measures directional alignment: vehicles with similar headings have small phase differences, whereas sharp heading conflicts generate larger offsets. The factor $1_{\left\{D_i\neq D_j\right\}}$ penalizes vehicles with different downstream exits, capturing diverging long-term intentions. The bearing term $arctan\left(\frac{y_j-y_i}{x_j-x_i}\right)-\psi i$ introduces spatial context without dominating behavioural cues. Consequently, side-by-side vehicles travelling in the same direction toward the same destination exhibit small phase differences and thus interfere constructively, whereas vehicles with conflicting destinations, heading misalignment, or opposite segment ordering accumulate larger phase offsets and tend to interfere destructively.

The resulting quantum-inspired kernel becomes:

$$\widetilde{K}\left(\chi_i,\chi_j\right)=K_{cl}\left(\chi_i,\chi_j\right)exp\left(i\beta\phi\left(X_i,X_j\right)\right),\beta\in\mathbb{R}. \tag{15}$$

Since $\left|\exp\left(i\beta\phi\left(X_i,X_j\right)\right)\right|=1$, this multiplication preserves the boundedness of the kernel; the Lipschitz continuity of ϕ ensures that $\widetilde{K}$ remains Lipschitz in X_i on the compact state space.

The corresponding aggregate mean-field interaction integrates individual pairwise influences:

$$W_t(X) = \int_{\chi} \widetilde{K}(\chi, \chi^j)\mu_t(d\chi^j) \tag{16}$$

Equation (16) embodies the quantum-inspired enhancement by combining neighbors' influences as complex vectors rather than scalars, introducing interference patterns naturally through complex vector addition. This mechanism automatically amplifies cooperative maneuvers without explicit coordination algorithms, as aligned vehicle trajectories reinforce each other's influence vectors. Conversely, conflicting maneuvers, such as one vehicle entering a merging zone while another exits, result in opposing vectors that partially cancel, thereby moderating conflicting steering commands and preventing dominance by contradictory cues.

This interference-driven approach enables spontaneous, self-organized traffic patterns, such as adaptive zipper merges, without centralized control. Unlike classical methods, where conflicting signals simply aggregate and potentially destabilize control commands, quantum-inspired interference maintains stable collective behavior, effectively resolving conflicts at high traffic densities. Consequently, the quantum-enhanced kernel, $\widetilde{K}$, captures not just interaction magnitudes but also their coherent structural relationships, significantly improving GMFG realism, solution stability, and overall system efficiency while maintaining the theoretical guarantees of existence and uniqueness.

3.3. Boundedness and Lipschitz regularity of kernel interaction function

All exponential and Bearing factors satisfy $0.3 \le D(\theta_{ij}) \le 1$ and $\exp(.) \le 1$ when arguments are nonnegative. The destination coupling weight satisfies $1 \le 1 + \kappa_0 1_{\{D_i = D_j\}} \le 1 + \kappa_0$. In the nudging term $N(X_i, X_j)$ and at any given longitudinal location, only one of the two ramp-type sums in $N(X_i, X_j)$ can be appreciable, hence, $|N(X_i, X_j)| \le 2\lambda$. Finally, the unit-modulus phase factor $|\exp(i\beta\phi(X_i, X_j))| = 1$ does not change the magnitude. Thus,

$$\left|\widetilde{K}(\chi_i, \chi_j)\right| \le (1+\kappa) + 2\lambda,$$

and integration against the probability measure μ_t following the Riesz–Markov representation theorem (Carmona & Delarue, 2018) yields to the uniform bound

$$|W_t(X)| \le (1+\kappa) + \lambda.2$$

Moreover, because each scalar factor and the phase function ϕ are Lipschitz (or can be arbitrarily well-approximated by Lipschitz maps) on the compact domain, there exists a constant $V > 0$ such that for any two states χ_1, χ_2, and any neighbor χ',

$$\left|\widetilde{K}(\chi_1, \chi') - \widetilde{K}(\chi_2, \chi')\right| \le L\|\chi_1 - \chi_2\|$$

Integrating against μ_t implies:

$$|W_t(\chi_1) - W_t(\chi_2)| \le L\|\chi_1 - \chi_2\|.$$

Since χ is compact, these properties guarantee the McKean–Vlasov operator built on W_t admits a unique fixed point and satisfies propagation of chaos, completing the regularity proof for the quantum-inspired, lane-free mean-field interaction kernel (Sznitman, 1991).

3.3.1. Pay-off or cost function

As in all game-theoretical approaches, defining a payoff or cost function is essential to systematically guide and evaluate agents' decisions, ensuring that each vehicle's chosen actions reflect rational trade-offs aligned with individual and collective goals. Thus, a cost functional for each vehicle is designed to balance four objectives control of:

1) acceleration and lateral steering to ensure energy efficiency and passenger comfort,
2) speed tracking to enforce adherence with a desired speed,
3) timely phase progression for smooth transition through merge/diverge region), and.
4) cooperative "coupling" with peers to synchronize driving behaviour when needed.

The cumulative cost over a fixed horizon $[0, T]$ for a representative vehicle is:

$$J[A(.); \mu(.), \eta(.)] = \mathbb{E}\left[\int_t^T L(X(t), A(t); W_t, \eta(t))dt + g(X(T))\right] \tag{17}$$

Where g is a penalty enforcing correct exit behaviour (Lasry & Lions 2007; Huang et al. 2006).

To render each term of the following equation dimensionless and interpretable on $[0, 1]$, we introduce characteristic scales A_{max} for acceleration, v_{max} for velocity, and $\Delta S_{max} = 4$ for the maximum discrete-phase deviation.

We define the running cost function as follows:

$$L(X(t), A(x(t)); W_t, \eta(t)) = \omega_u \frac{||A||^2}{A_{max}^2} + \omega_v \frac{||v - v_{ref}(S)||^2}{v_{max}^2} + \omega_s \frac{||S - S_{target}(O, D)||^2}{\Delta S_{max}^2} \tag{18}$$

$$- \omega_e E_{norm}(X) + \omega_c C_{gap}(X)$$

With convex weights $\omega_u, \omega_v, \omega_s, \omega_e \geq 0$, summing to one. In this paper, these weights were determined through sensitivity analysis to reflect the relative importance of each cost component, but the formulation can be extended to allow them to be learned or adjusted dynamically during training to adapt to varying traffic conditions or learning phases. The structure of the running cost follows the classical convex decomposition widely used in game theory and optimal driving formulations (Lasry & Lions, 2007; Huang et al., 2006; Mo et al., 2024). Each term corresponds to a physically meaningful driving objective and ensures that the resulting HJB minimization remains convex, interpretable, and numerically well-posed. This formulation is fully consistent with established microscopic and macroscopic traffic-flow theory, where smooth accelerations, adherence to desired speeds, and anticipatory cooperative behaviour constitute the fundamental elements of safe and efficient driving (Treiber & Helbing, 2002). The specific functional forms were therefore chosen not ad hoc, but because they encode: comfort and energy efficiency via acceleration regularization, traffic-efficiency via speed-tracking penalties, correct routing through phase-progression costs, and coordinated multi-vehicle behaviour through a coherence-alignment term, and safety through a speed-dependent safety gap term inspired by Yanumula el al. (2023) and Dabestani el al. (2025). Each term of Equation (18) plays the following role:

- Control effort $\frac{||A||^2}{A_{max}^2}$: penalizes aggressive maneuvers, encouraging energy-efficient and passenger-comfortable trajectories (Bain & Crisan 2008).
- Speed tracking $\frac{\left|\left|v - v_{ref}(S)\right|\right|^2}{v_{max}^2}$ enforces adherence to a region-dependent reference speed $v_{ref}(S)$, such as slower velocities for safe driving in merging zones.
- Phase-alignment $\frac{\left|\left|S - S_{target}(O,D)\right|\right|^2}{\Delta S_{max}^2}$ guides vehicles through their prescribed sequence of phases, merge, cruise, or diverge, so that they reach their intended exit in proper order without back-tracking.
- Coherence reward $-E_{norm}(X)$ Rewards cooperative synchronization with surrounding traffic via quantum-aligned steering. This term arising from quantum-inspired alignment between an agent's maneuver and the collective behavior is defined by:

$$E_{norm}(X) = (\Re\{\overline{W(X)}\psi(X)\} + 1)/2 \in [0,\ 1], \tag{19}$$

This equation quantifies how well the agent's manoeuvre is behaviourally phase-aligned with the aggregate population influence $W(X)$ defined in Equation (16). The complex exponential term in the kernel encodes coherence of behaviour, through heading alignment, destination compatibility, and geometric distance, remains exclusively in the magnitude K_{cl}. The inner product $W(X)\overline{\psi(X)}$, therefore measures whether the agent's intended motion reinforces or opposes the dominant behavioural pattern of surrounding traffic. Its real part captures the degree of constructive versus destructive interference, and the normalization $(\Re(\cdot)+1)/2$ maps this value from $[-1,1]$ into $[0,1]$. Values near 1 indicate strong behavioural alignment with nearby traffic, whereas values near 0 indicate misalignment or conflicting intentions. This makes the alignment reward E_{norm} a bounded and interpretable measure of cooperative synchrony within the quantum-inspired GMFG framework.

- To explicitly address collision avoidance and reduce reliance on emergency interventions, we augment the running cost with a smooth speed-dependent safety-gap penalty $C_{gap}(X)$. This term discourages vehicles from entering unsafe proximity configurations by penalizing intrusions into a speed-dependent safety envelope around neighboring vehicles. This allows safety considerations to be embedded directly into the equilibrium optimization, rather than enforced through post-hoc emergency actions. Emergency interventions act as a conservative safety-override layer and do not correspond to collision events.

For each vehicle i and neighboring vehicle j, safety is modeled through a smooth ellipsoidal keep-out region defined in the ego-vehicle reference frame (Yanumula el al., 2023; Dabestani el al., 2025). Let $\left(\Delta x_{ij}, \Delta y_{ij}\right)$ denote the longitudinal and lateral relative positions of vehicle j with respect to vehicle i. The normalized intrusion metric is defined as $g_{ij}(X) = \left(\frac{\Delta x_{ij} - \delta_{ij}}{d_{1,ij}}\right)^2 + \left(\frac{\Delta y_{ij}}{d_{2,ij}}\right)^2$.

The longitudinal and lateral semi-axes of the ellipsoid scale with vehicle speed according to $d_{1,ij} = l_i + \omega_1|v_i| + \omega_1|v_j|$ and $d_{2,ij} = w_i + \omega_2|v_i| + \omega_2|v_j|$, respectively. l_i and w_i account for the physical dimensions of the ego and neighboring vehicles, and ω_1, ω_2 represent longitudinal and lateral time-gap parameters, respectively. The ellipsoid center is shifted longitudinally by $\delta_{ij} = -\frac{\omega_1}{2}\left(|v_i| - |v_j|\right)$, allowing asymmetric front–back safety gaps that depend on relative speed. This formulation ensures that safety margins expand naturally with velocity.

A smooth safety-gap penalty is then defined as

$$C_{\text{gap}}(X) = \sum_{j \in \mathcal{N}_i} \log\left(1 + \exp\left(\kappa\left(1 - g_{ij}(X)\right)\right)\right),$$

where $\kappa > 0$ controls the sharpness of the penalty near the safety boundary. When $g_{ij}(X) > 1$, the vehicle lies outside the safety envelope and the penalty is negligible; when $g_{ij}(X) < 1$, the penalty increases smoothly as the vehicle approaches or intrudes into the unsafe region. This barrier-like but fully differentiable formulation enables anticipatory collision avoidance within the GMFG

optimization.

Altogether, this cost structure enables each vehicle to independently optimize its trajectory while fostering coordinated system-wide behaviours that can improve throughput and reduce turbulence (Meyer, 1999; Dey et al. 2024). This formulation is consistent with the quantum-inspired GMFG framework.

While the present study assumes homogeneous vehicles to focus on the core GMFG formulation, the framework can be naturally extended to heterogeneous traffic with different vehicle sizes, dynamics, and maximum speeds (e.g., cars and trucks). This can be achieved by introducing class-specific state constraints, and cost parameters within a multi-population mean-field game formulation, while preserving decentralized execution based on locally available information.

In summary, Equation (18) formulates the running cost used for control optimization, utilizing selected components of the physical state to quantify the performance objectives. This stands in contrast to Equation (5), which explicitly describes how the vehicle's physical state evolves over time subject to kinematic constraints. The difference in variables between these two equations reflects the fundamental distinction between cost evaluation (optimizing for desirable behaviour) and motion modeling (enforcing physical feasibility), rather than the introduction of additional vehicle states.

3.3.2. Adaptive coherence mechanism

The coherence mechanism creates an adaptive response to evolving traffic conditions. It enables the system to dynamically strengthen cooperative behaviours when vehicles need to act collectively while allowing cooperation to diminish when vehicles operate independently, creating an adaptive response to emerging traffic conditions and collective driving behaviours. Thus, rather than treated as a static parameter, the coherence-gain τ is endowed with its own dynamics (ω_e in Equation (19) that capture the evolving degree of cooperative coherence. $\tau(t)$ is the scalar weight that modulates how strongly each vehicle's running cost is influenced by the population's collective phase coherence. The learning-style update $\tau(t)$ is given by the linear ordinary differential equation (ODE):

$$\dot{\tau}(t) = \kappa C(t) - \delta\tau(t), \tag{20}$$

Where $C(t)$ is the global phase coherence defined by Equation (21):

$$C(t) = |<\psi, \mu_t>| = \left| \int_{\chi} \psi(X)\mu_t(dX) \right|, \tag{21}$$

Where μ_t is the population distribution at time t. $C(t)$ aggregates phases *globally* into a single coherence metric. When all vehicles share the same φ, their complex phases add constructively and $C(t) \approx 1$; if phases are uniformly dispersed, cancellation yields $C(t) \approx 0$ (Helbing, 2001).

In Equation (20), the reinforcement rate $\kappa > 0$ determines how rapidly τ increases in response to emerging synchronization, mirroring positive social feedback in driving behaviour. The decay rate δ provides a "forgetting" mechanism that dampens τ when coherence diminishes, thereby ensuring stability under changing traffic conditions (Ioannou & Sun, 2012). A larger τ amplifies the coherence reward in the running cost, encouraging each vehicle to align its phase with the population average. These models human-like behaviors such as alternating zipper merges (Jain et al., 2011) and anticipatory lateral movements in response to merging neighbors (Treiber & Helbing, 2002).

Because ψ is bounded and Lipschitz and $C(t)$ is a linear function of the compactly supported measure μ_t , the right-hand side of the ODE is Lipschitz in τ and μ_t . Consequently, the joint forward–backward McKean–Vlasov– GMFG system remains well-posed, preserving compactness and guaranteeing the existence and uniqueness of the equilibrium (Carmona & Delarue, 2018).

In contrast to classical lane-free trajectory planning frameworks that focus primarily on ego-centric navigation, the running cost in Equation (18) incorporates the population density directly through the mean-field interaction amplitude $W(X)$ and the adaptive coherence mechanism $\tau(t)$. This ensures that each vehicle's microscopic control action depends not only on its local state but also on the evolving macroscopic density field. The inclusion of segment-based kernels, phase-alignment terms, and coherence rewards creates a direct micro–macro feedback mechanism: vehicles respond to global congestion patterns, while their collective decisions determine the evolution of traffic. This explicit coupling is foundational to GMFG theory and distinguishes the present formulation from earlier LFT models, which typically rely on local interaction rules.

3.3.3. GMFG equilibrium conditions

A Graphon Mean-Field Game equilibrium consists of a triplet $(u,\ \mu,\ \tau)$ and a feedback control $A^*(x,t)$ that satisfy a coupled system of equations representing optimality, population evolution, and adaptive coherence dynamics. This equilibrium describes a dynamic consistency between agents' optimal behaviours and the evolving traffic environment over the entire time horizon. The time dependence of the optimal control $A^*(x,t)$ in Eq. (23) reflects changing vehicle states and local traffic interactions. The control represents a feedback policy evaluated at time t using the vehicle's current state and nearby traffic conditions. The policy is learned offline within the GMFG framework and has the same functional form for all vehicles. During operation, each vehicle applies this policy independently using locally available information, ensuring decentralized execution. An equilibrium therefore ensures that each vehicle acts optimally given the collective behaviour, and that the traffic distribution and coherence strength evolve in a way that is coherent with those individual decisions. Solving for equilibrium continuously in time avoids oscillatory or myopic behaviours and provides stability and predictability across the system. It also ensures analytical tractability, as guaranteed under McKean–Vlasov theory. The equilibrium satisfies the following equation:

1. Hamilton–Jacobi–Bellman equation (optimality)
2. The value function $u(x,t)$ solves, backward in time:

$$-\partial_t u(x,t) = min_{a\in A}\{L(x,a;W_t(x),\eta(t)) + \nabla_x u(x,t).f(x,a)\}, u(x,T) = g(x) \tag{22}$$

The minimizer yields the feedback law (*Huangetal.*, 2006):

$$A^*(x) = argmin_a\{L + \nabla_x u.f\} \tag{23}$$

3. Fokker–Planck (continuity) equation
4. The population density μ_t evolves forward in time as:

$$\partial_t \mu_t = \nabla_x.(f(x,A^*(x))\mu_t(x)) = 0, \mu_0 = \mu_{init} \tag{24}$$

Here $f(x,a)$ is the deterministic drift from the state-transition map (Lasry & Lions 2007).
5. Coherence-gain evolution

$$\dot{\tau}(t) = \kappa C(t) - \delta\tau(t), \tau(0) = \tau_0.$$

Under the Lipschitz and compactness conditions, and standard McKean–Vlasov theory, this forward–backward–adaptive system admits a unique solution (Carmona & Delarue 2018; Caines & Huang 2021).

The equilibrium conditions in Equations (22)–(24) embody a closed-loop interaction between microscopic decisions and the evolution of macroscopic flow. Each agent's best response depends on the macroscopic density through $W(X)$ and $\tau(t)$, and the resulting ensemble of optimal trajectories drives the population distribution forward via the FP equation. This produces a bidirectional coupling in which microscopic controls internalize macroscopic traffic patterns, and those patterns in turn evolve according to the aggregate effect of optimal individual actions. This explicit micro–macro feedback loop clarifies how the proposed microscopic formulation drives the macroscopic phenomena observed in the simulation results.

The HJB solution is approximated offline using an actor–critic RL scheme within the GMFG framework, starting from standard network initialization and trained on large-scale traffic simulations until convergence. The resulting feedback policy is then fixed and evaluated locally by each vehicle during operation.

It is worth noting that the proposed GMFG formulation is fully deterministic and does not incorporate explicit stochastic noise in sensing, communication, or vehicle dynamics, as the primary objective of this work is to study interaction structure and equilibrium behaviour rather than robustness to random perturbations.

3.3.4. Solution algorithm: reinforcement-learning–based solver

In this paper, the high dimensionality of the lane-free GMFG state, comprising four continuous coordinates (x, y, v_x, v_y), three discrete regime labels (S,O,D), and the scalar coherence gain τ, renders classical partial differential equation solvers intractable. The associated GMFG equilibrium corresponds to a nonlinear HJB equation coupled with a McKean–Vlasov Fokker–Planck equation defined on a seven-dimensional hybrid state space. Grid-based numerical schemes for such HJB systems suffer from exponential growth in complexity (the "curse of dimensionality"), making them computationally infeasible even under coarse discretization (Fleming & Soner, 2006). Furthermore, viscosity-solution approaches (Crandall & Lions, 1983) require full discretization of the state domain and therefore cannot be applied to models involving nonlinear bicycle dynamics, complex-valued interaction kernels, and adaptive coherence states. Recent advances show that deep neural networks can approximate high-dimensional viscosity solutions of HJB equations (Han et al., 2018), and that actor–critic architectures provide consistent numerical approximations for continuous-action MFG and McKean–Vlasov control problems (Perolat et al., 2017; Carmona and Laurière, 2021). For these reasons, an actor–-critic reinforcement-learning framework is adopted as a numerical solver for the GMFG equilibrium rather than as a heuristic control design.

In this formulation, the reward is not artificially shaped; it is the exact running cost $L(\chi,u,\tau)$ defined in Equation (18), guaranteeing direct correspondence between RL optimization and the GMFG objective functional. The actor–critic method uses two feed-forward neural networks: the critic approximates the value function $u_\theta(X,t)$, while the actor approximates the control minimizing the Hamiltonian by producing the mean accelerations $\left(\mu_x, \mu_y\right)$ and a diagonal covariance matrix scaled to enforce the control constraint $|A| \le A_{max}$. Each network receives the four continuous state variables, a 16-dimensional embedding of each discrete label (S,O,D), and the normalized coherence gain $\tau \in [0,1]$. The continuous inputs are processed through two hidden layers of 128 ReLU units, a standard configuration shown to balance expressive power and training stability in high-dimensional control tasks (Han et al., 2018).

Approximation errors arise from the temporal discretization, the neural-network function approximation, the finite-N estimation of the mean-field amplitude, the stochastic temporal-difference learning, and the finite-population deviation from the GMFG limit. These errors are monitored through TD-error convergence, independent validation rollouts, and policy-stability assessments. GPU acceleration is used solely to parallelize the forward passes and neighbor-averaging computations (Equation (25) across thousands of agents, providing meaningful reductions in wall-clock time but without implying any theoretical complexity improvement. The resulting RL-based controller should therefore be interpreted as a tractable high-dimensional numerical approximation to the GMFG equilibrium, with an exact PDE-based benchmark solution identified as an important direction for future work.

The critic network $u_\theta(X,t)$ outputs a single scalar estimate of the value function, and the actor network $\pi_\theta(X,t)$ produces the mean accelerations $\left(\mu_x, \mu_y\right)$ and diagonal covariance matrix, parameterized to enforce the control constraints $||A|| \le A_{max}$.

Training is performed in synchronous mini-batches of $M = 1024$ simulation entities, a batch size chosen to balance gradient variance against GPU memory constraints in comparable multi-agent RL benchmarks (Perolat et al., 2017). Each simulation run consists of a time step $\Delta t = 0.2s$, which is small enough to resolve fast merging dynamics but large enough to keep rollouts computationally tractable. At each step, the actor samples an action $A_i \mathcal{N}(\mu_i, \sum)$ which is applied to the deterministic point-mass dynamics. After propagation, the mean-field interaction amplitude is determined as:

$$W_t(X_i) \approx \frac{1}{K} \sum_{j \in \mathcal{N}_K(i)} \widetilde{K}(\chi_i, \chi_j) \tag{25}$$

is estimated by averaging over the $K = 50$ nearest neighbors in Euclidean (x, y)-space, a neighborhood size chosen to minimize estimator bias while controlling variance in large-scale simulations. The phase observable $\psi(X_i) = e^{i\varphi_i}$ is computed from each vehicle's normalized merge progress φ_i, and these quantities feed into the running cost $L(X_i, A_i)$. The global coherence:

$$C = \left| \frac{1}{M} \sum_i \psi(X) \right| \tag{26}$$

is used to update the coherence gain via a forward Euler integration of the ODE $\dot{\tau} = \kappa C - \delta\tau$ with $\kappa = \delta = 0.5\,s^{-1}$. These symmetric rates ensure that τ responds on a time scale of approximately two seconds, matching human reaction times and allowing timely but not overly oscillatory adaptation (Ioannou & Sun, 2012).

For the critic update, one-step temporal-difference learning is used with discount factor $\gamma = 0.99$, which retains long-horizon reward information without numerical instability. The target for each transition $(X_i, A_i, L_i, X_{i'})$, is

$$y_i = L_i + \gamma u_\theta(X_i, t + \Delta t \tag{27}$$

and the critic parameters θ are optimized by minimizing the mean squared Bellman error using the Adam optimizer at a learning rate of 3×10^{-4}, a standard choice in modern RL benchmarks for its robustness to sparse or noisy gradients.

Simultaneously, the actor update uses the deterministic policy gradient theorem, subject to a trust-region constraint that limits the Kullback–Leibler divergence between successive policies to less than 10^{-2} per update; this ensures stable policy improvement without large, destabilizing parameter shifts.

Each training epoch consists of 200 rollout steps followed by 50 gradient-descent updates, a schedule that provides sufficient on-policy data to update both networks effectively. Training continues for up to 10,000 episodes or until convergence criteria are met, namely, an average TD-error below 10^{-2} and less than 1% change in τ over 100 episodes. This configuration retains the convergence guarantees and sample efficiency of modern actor–critic methods (Perolat et al., 2017; Han et al., 2018) while faithfully approximating the coupled HJB–FP– coherence equilibrium of the quantum-inspired lane-free traffic model.

3.4. String stability analysis

String stability in our context means that any small disturbance, whether a slight breaking in the lead vehicle or a minor lateral deviation, is designed not to grow as it propagates through the stream of CAVs. In other words, if one vehicle applies a brief deceleration or lateral shift, string stability guarantees that its followers react in a way that dampens, rather than amplifies, the perturbation. In the lateral dimension, the nudging-kernel consensus law (Section 3.1.3) yields nonlocal feedback whose Fourier-mode analysis shows that every spatial frequency obeys a neutral or decaying response (no mode ever has a positive growth rate), so small steering errors cannot propagate into large weaving oscillations. Longitudinally, the exponentially decaying headway kernel similarly ensures that velocity fluctuations ahead are attenuated downstream.

The stability analysis in this section is carried out for the complete hybrid system, including all region-dependent operating conditions. Although the corridor contains five physical regimes, the controller itself is not segmented: all vehicles apply the same GMFG-derived feedback law, conditioned on their full state, including the region label. As a result, the lateral and longitudinal perturbation dynamics and the resulting stability guarantees hold uniformly across pre-merge, merge, cruising, diverging, and post-diverge zones. The GMFG framework solves a single Hamilton–Jacobi–Bellman equation over the full state space, and the region label simply modulates the parameters of the dynamics and objective without creating independent controllers. Thus, the entire corridor is optimized and stabilized as one coupled system rather than as a collection of separate control problems.

The following proofs demonstrate that the GMFG-derived control laws maintain both lateral and longitudinal string stability, preventing shock waves and stop–and–go chains in a lane-free highway environment. Section 3.2.1 presents the derivation of lateral string stability, while Section 3.2.2 presents the longitudinal counterpart.

3.4.1. Lateral string stability

The key idea is that, under the formulated nudging kernel (Equation (12), any small lateral perturbation is governed by a convolutional feedback law whose frequency response never amplifies.

In the present lane-free traffic, lateral string stability does not imply an infinite number of side-by-side vehicles or lanes. Instead, the lateral coordinate represents the cross-stream distribution of vehicles inside the finite merge and diverge envelope, where ramp vehicles and mainline vehicles interact within a bounded lateral width. As is standard in continuum and string-stability analyses, we use an infinite-domain harmonic expansion as a mathematical device for characterizing worst-case amplification; the resulting non-

amplification conditions remain valid for long but finite corridors by truncation.

Let's examine the one-dimensional lateral dynamics induced by the nudging field N, consider the following steps:

4. Transition from discrete sum to continuum integral

Recall that in the full kernel, each neighbor j contributes a term of the form (Equation (12):

$$N_{ij} = exp(-|d(x_i)|/\eta).exp\left(-\left|y_j - y_i\right|/\eta_\perp\right).sign(d(x_i)),$$

Assume small lateral deviations $\delta y_i = y_j - y_i$ are close to 0, and the vehicle is being pulled in the positive lateral direction. This allows us to apply a first-order Taylor expansion around $\delta y_i \to 0$ in the following expression:

$$exp\left(-\left|y_j - y_i\right|/\eta_\perp\right).sign(d(x_i)) \approx \left(exp(0) - \delta y_i.\left(\frac{1}{\eta_\perp}exp\left(-\frac{|\delta y_i|}{\eta_\perp}\right)\right)_{\delta y_i = 0}\right).1 \approx 1 - \frac{\delta y_i}{\eta_\perp} \approx 1 - \frac{\delta y_j}{\eta_\perp} \Rightarrow N_{ij} \approx -w_{i,j}.\delta y_j$$

Where, $w_{i,j} = \exp\left(-\frac{|x_j - x_i|}{\eta}\right)$ collects the leading longitudinal decay.

The expression

$$U_i(y) = \frac{\lambda}{2}\sum_j w_{ij}\left(y_j(t) - y_i(t)\right)^2 \tag{28}$$

originates in classical potential-field theory as a convenient and smooth way to encode lateral alignment or "consensus" behaviour among agents. By choosing a quadratic form $\left(y_j(t) - y_i(t)\right)^2$, larger deviations are penalized more heavily while retaining differentiability and yielding linear feedback upon differentiation. The weights $w_{i,j}$ reflect the fact that vehicles closer in the longitudinal direction should exert stronger lateral influence, with η setting the decay scale. The prefactor $\lambda > 0$ scales the overall strength of this nudging behaviour, and the conventional $1/2$ normalization ensures that the resulting lateral force, given by the negative gradient $-\frac{\partial U_i}{\partial y_i}$, simplifies to:

$$\ddot{y}_i(t) = -\frac{\partial U_i}{\partial y_i} = \lambda\sum_j w_{ij}\left[y_j(t) - y_i(t)\right], \tag{29}$$

a classic consensus or alignment law. In this way, U_i defines a smooth potential energy landscape whose minimization precisely drives each vehicle's lateral acceleration toward the weighted average lateral position of its neighbors, seamlessly extending established flocking and social-force models to the lane-free traffic setting.

For a large, regularly spaced platoon of vehicles at density ρ_0, one can pass to the continuum limit by introducing the field

$$y(x,t) \approx y_j(t) when x_i \approx x,$$

and replacing the discrete sum with a non-local continuum integral weighted by ρ_0 :

$$\sum_j w_{ij}\left[y_j - y_i\right] \to \rho_0 \int_{-\infty}^{\infty} e^{-|r|/\eta}[y(x+r,t) - y(x,t)]dr \tag{30}$$

Equation (27) rests on the mean-field or continuum limit commonly employed in large-scale agent systems. In the discrete form, each vehicle i experiences a weighted consensus pull toward its neighbors' lateral positions $y_j - y_i$, with weights decaying exponentially in longitudinal separation. When the vehicle density is high and sampling becomes effectively continuous, the index-based sum can be interpreted as a Riemann sum over offset $r = x_j - x_i$. Under the assumption of a uniform longitudinal density ρ_0 (vehicles per unit length), the sum $\sum_j f(x_j) \approx \rho_0 \int f(x+r)dr$ transforms the discrete interactions into the convolution-type integral above. Here $y_j(t) \mapsto y(x+r,t)$ and $y_i(t) \mapsto y(x,t)$, so that the local consensus update becomes a spatially nonlocal feedback term: each point x "senses" the lateral deviations in its entire neighborhood, exponentially weighted by the kernel $e^{-|r|/\eta}$. This continuum formulation not only simplifies the analysis by enabling Fourier-mode stability proofs and PDE-based treatments, but also captures the same smooth, proximity-weighted alignment dynamics originally encoded by the discrete sum.

4.1. Linearization and Fourier transform

Starting from the nonlinear continuum nudging model in Equation (27), the first step is to examine small perturbations about the uniform equilibrium $y(x, t) = \bar{y}$. By looking at small deviations, $y(x, t) = \bar{y} + \delta y(x,t)$. Substituting into the integral, the constant, the constant $\bar{y}$ cancels, yielding the linearized equation:

$$\partial_t^2 \delta y(x,t) = \lambda.\rho_0 \int_{-\infty}^{\infty} e^{-|r|/\eta}[\delta y(x+r,t) - \delta y(x,t)]dr \tag{31}$$

This form isolates how deviations δy evolve under the nonlocal, exponentially weighted feedback. To analyze stability and

dispersion, one then applies the spatial Fourier transform:

$$\widehat{\delta y}(k,t) = \int_{-\infty}^{\infty} \delta y(x,t) e^{-ikx} dx,$$

using the property that convolution in x becomes multiplication in the wavenumber k. Under this transform, the Equation (28) becomes:

$$\partial_t^2 \delta y(x,t) = \lambda.\rho_0 \left(\widehat{e^{-\frac{|r|}{\eta}}}(k) - \widehat{e^{-\frac{|r|}{\eta}}}(0) \right) \widehat{\delta y}(k,t), \tag{32}$$

Where the Fourier transform of the Laplace kernel is:

$$\widehat{e^{-\frac{|r|}{\eta}}}(k) = \int_{-\infty}^{\infty} e^{-\frac{|r|}{\eta}} e^{-ikx} dr = \frac{2\eta}{1+(\eta k)^2}$$

Subtracting the zero-mode $\left(\widehat{e^{-\frac{|r|}{\eta}}}(0) \right) = 2\eta$ ensures that a uniform ($k = 0$) perturbation produces no net acceleration. The result is a decoupled ordinary differential equation (ODE) in time for each Fourier mode, k:

$$\partial_t^2 \widehat{\delta y}(k,t) = \lambda.\rho_0 \left(\frac{2\eta}{1+(\eta k)^2} - 2\eta \right) \widehat{\delta y}(k,t), \tag{33}$$

which can be studied analytically to determine whether and how small lateral deviations decay or oscillate as they propagate through the traffic flow.

4.2. Harmonic-oscillator form and non-amplification

A little algebra shows the coefficient $\left(\frac{2\eta}{1+(\eta k)^2} - 2\eta \right)$ is negative (or zero) for every real k, so we may write.

$$\partial_t^2 \widehat{\delta y} = -\omega_k^2 \widehat{\delta y}, \omega_k^2 = 2\lambda.\rho_0.\eta \frac{(\eta k)^2}{1+(\eta k)^2} \geq 0. \tag{34}$$

This is exactly the simple harmonic-oscillator equation (possibly with zero frequency for $k = 0$). Crucially, there is no regime in which ω_k^2 becomes negative, so no mode grows exponentially in time.

This exercise shows the mathematical proof of string stability in the lateral direction. Hence, lateral perturbations cannot amplify; they remain bounded or decay (once damping is added).

4.2.1. Longitudinal string stability

A parallel derivation can be carried out for longitudinal velocity perturbations under an interaction law of the form:

$$\dot{v}_i = \sum_j K(r).(x_j - x_i)\left[v_j - v_i\right], \tag{35}$$

Where $K(r) = \exp(-\alpha |r|)$. This longitudinal-velocity law arises by the same consensus-potential reasoning that underlies the lateral nudging model, now applied to speed mismatches rather than lateral offsets. In the continuum limit, small velocity deviations $\delta v(x,t)$ satisfy:

$$\partial_t \delta v(x,t) = \rho_0 \int K(r)[\delta v(x+r,t) - \delta v(x,t)]dr, \tag{36}$$

and Fourier analysis again leads to a multiplication $\widehat{K}(k) - \widehat{K}(0)$, which is non-positive for all k when K is a positive, symmetric, decaying function. One finds:

$$\partial_t \widehat{\delta v}(k,t) = -G(k)\widehat{\delta v}(k,t), \tag{37}$$

With $G(k) \geq 0$ for all k. The solution $\widehat{\delta v}(k,t) = \exp(-G(k)t).\widehat{\delta v}(k,0)$ therefore decays in time, proving longitudinal string stability: small speed perturbations die out rather than amplify downstream.

4.2.2. Two-dimensional string stability

In this section, we derive the two-dimensional linear perturbation equations required for the joint longitudinal–lateral string-stability analysis. We begin by linearizing the kinematic bicycle model around a homogeneous freeway equilibrium and then show how the resulting perturbation dynamics decompose into longitudinal and lateral components. This structure allows the spectrum of

2D disturbances to be obtained by the superposition of longitudinal and lateral Fourier modes.

4.2.3. Linearization of the kinematic bicycle model

Each vehicle follows the nonlinear kinematic bicycle model in the corridor-aligned global frame,

$$\dot{x} = v\cos\psi, \dot{y} = v\sin\psi, \dot{v} = a, \dot{\psi} = \frac{v}{L}\tan\delta \tag{38}$$

A physically meaningful freeway equilibrium is characterised by uniform motion along the road direction:

$$v_0 = \text{constant}, \psi_0 = 0, \delta_0 = 0, a_0 = 0 \tag{39}$$

Although the system is embedded in two spatial dimensions, the equilibrium trajectory is one-dimensional, and all lateral motion arises from perturbations. This is standard in vehicular and platooning stability analysis.

Let $\tilde{x}, \tilde{y}, \tilde{v}, \tilde{\psi}, \tilde{a}, \tilde{\delta}$ denote deviations from this equilibrium. Under small-angle approximations ($|\tilde{\psi}| \ll 1, |\tilde{\delta}| \ll 1$), linearization of Equation (38) yields:

$$\dot{\tilde{x}} = \tilde{v}, \dot{\tilde{v}} = \tilde{a}, \dot{\tilde{y}} = v_0\tilde{\psi}, \dot{\tilde{\psi}} = \frac{v_0}{L}\tilde{\delta} \tag{40}$$

Differentiating t $\dot{\tilde{y}}$ and substituting $\dot{\tilde{\psi}}$ gives the second-order lateral perturbation dynamics:

$$\ddot{\tilde{y}} = \frac{v_0^2}{L}\tilde{\delta} \tag{41}$$

Thus, in the perturbation regime relevant for string stability, longitudinal and lateral motions satisfy:

$$\ddot{\tilde{x}} = \tilde{a}, \ddot{\tilde{y}} = \frac{v_0^2}{L}\tilde{\delta} \tag{42}$$

These linearized relations underpin all subsequent continuum-mode and spectral analyses.

4.2.4. Linearized interaction forces and operator decomposition

The GMFG controller induces cooperative longitudinal and lateral interactions via linearized kernel forces. Let $\tilde{x}(x, y, t)$ and $\tilde{y}(x, y, t)$ denote perturbation fields. The longitudinal and lateral control inputs generated by the linearised kernel take the form:

$$\tilde{a}(x, y, t) = \int K_x(x - x', y - y')(\tilde{x}(x', y', t) - \tilde{x}(x, y, t))\, dx'dy' \tag{43}$$

$$\tilde{\delta}(x, y, t) = \int K_y(x - x', y - y')(\tilde{y}(x', y', t) - \tilde{y}(x, y, t))\, dx'dy' \tag{44}$$

Substituting *Equation (43) and Equation (44)* into *Equation (42)* yields the second-order perturbation system:

$$\ddot{\tilde{x}} = \int K_x(\,\cdot\,)(\tilde{x}' - \tilde{x})\, dx'dy', \ddot{\tilde{y}} = \frac{v_0^2}{L}\int K_y(\,\cdot\,)(\tilde{y}' - \tilde{y})\, dx'dy' \tag{45}$$

Define the perturbation vector field as:

$$z(x, y, t) = \begin{bmatrix} \tilde{x}(x, y, t) \\ \tilde{y}(x, y, t) \end{bmatrix} \tag{46}$$

Define the longitudinal operator $\mathscr{L}_x$ and lateral operator $\mathscr{L}_y$ as:

$$\mathscr{L}_x z = \begin{bmatrix} \int K_x(\,\cdot\,)(\tilde{x}' - \tilde{x})dx'dy' \\ 0 \end{bmatrix}, \mathscr{L}_y z = \begin{bmatrix} 0 \\ \frac{v_0^2}{L}\int K_y(\,\cdot\,)(\tilde{y}' - \tilde{y})dx'dy' \end{bmatrix} \tag{47}$$

The full 2D perturbation evolution is therefore:

$$\partial_{tt} z = \mathscr{L} z, \mathscr{L} = \mathscr{L}_x + \mathscr{L}_y \tag{48}$$

Thus, longitudinal and lateral perturbations are decoupled in the linear regime, and the overall operator is the sum of their respective linear generators.

4.3. Spectral decomposition and two-dimensional growth rates

Because the kernels depend only on spatial differences, $K_x(x - x', y - y'), K_y(x - x', y - y')$, the operators $\mathscr{L}_x$ and $\mathscr{L}_y$ are convolution operators and hence diagonalised by Fourier modes.

Consider a perturbation of the form:

$$z(x,y,t) = \widehat{z}\, e^{i(k_x x + k_y y) + \lambda t} \tag{49}$$

Substituting *Equation (49)* into *Equation (48)* yields the modal growth rates

$$\lambda(k_x, k_y) = \lambda_x(k_x) + \lambda_y(k_y) \tag{50}$$

where λ_x and λ_y are the longitudinal and lateral eigenvalues induced respectively by $\mathscr{L}_x$ and $\mathscr{L}_y$. The string-stability conditions derived earlier ensure that $\lambda_x(k_x) \le 0$ and $\lambda_y(k_y) \le 0$, for all admissible wavenumbers. Consequently, $\lambda(k_x, k_y) \le 0$, and the total perturbation energy:

$$E(t) = \iint \left\| z(x,y,t) \right\|^2 dx\, dy \tag{52}$$

is non-increasing over time.

In summary, these results establish that the coupled HJB–FP– coherence dynamics underpinning the GMFG equilibrium admit no regime in which microscopic perturbations amplify downstream and laterally, thereby ensuring absorption of shockwave.

5. Simulation and results

To validate the quantum-inspired GMFG methodology, a comprehensive microsimulation framework was developed. The simulation environment models a 6-kilometer lane-free highway f 10.2 m width and three on-ramps positioned at longitudinal coordinates 1,000 m, 3000 m, and 5000 m, and three off-ramps located at 1350 m, 3350 m, and 5350 m. This multiple on-ramps off-ramps configuration creates challenging weaving sections to examine the performance of developed framework.

Traffic demand follows a trapezoidal profile that represents typical peak period traffic highway loading. The demand, loaded following a log-normal distribution, increases linearly from 10,000 vehicles/hour to 20,000 vehicles/hour over the first 20 min of simulation, maintains the peak demand level for 20 min, then decreases linearly to 10,000 vehicles/hour over the final 20 min. Origin-destination patterns are configured to allocate 8% of total demand to each on-ramp entry point and 8% of exits to each off-ramp, with the remaining 76% of traffic utilizing the mainline corridor from upstream to downstream. Free-flow speed is set at 100 km/h on the entire corridor except 80 km/h in merging and diverging zones.

The model weights were selected through exploratory testing of multiple parameter combinations, and the configuration yielding the highest sustained throughput while maintaining stable and collision-free operation was adopted for the reported simulations. Preliminary sensitivity tests further indicated that moderate variations of these weights do not alter the qualitative macroscopic behaviors (less than 10% change), primarily affecting the strength and convergence rate of coordination.

Vehicle control employs a RL −based policy for longitudinal acceleration, while the Intelligent Driver Model (IDM) is used as a safety reference to guide nominal longitudinal spacing. In the LFT environment, the IDM is adapted by defining a virtual leading vehicle, identified at each time step as the nearest vehicle ahead within a local heading-aligned interaction region. The longitudinal gap and relative speed required by the IDM are computed by projecting the relative position and velocity of this interaction partner onto the ego vehicle's direction of motion, thereby preserving the longitudinal car-following interpretation of the IDM without relying on lane structure. During training, the RL objective is regularized using the IDM-recommended acceleration to encourage safe longitudinal spacing, while hard minimum-distance constraints strictly enforce collision avoidance. Lateral safety is enforced independently through a minimum lateral clearance constraint and strict road-boundary constraints, ensuring safe vehicle separation in continuous space without relying on lane-based risk measures.

A collision risk metric is computed using time-to-collision (TTC) analysis, where $risk = \max\left(0,\, 1 - \frac{TTC}{5.0}\right)$ for approaching vehicle pairs, and is bounded between 0 (safe) to 1 (imminent collision) (Mullakkal-Babu et al., 2017). When collision risk exceeds 0.7 or following distance drops below 20 m, the system transitions to pure IDM control with emergency deceleration capability extending to -6.0 m/s^2. The IDM implementation employs standard parameters: desired speed v_0=100 km/h, time headway T = 1.5 s, maximum acceleration a = 2.0 m/s^2, comfortable deceleration b = -3.0 m/s^2, minimum spacing s_0=2.0 m, and acceleration exponent $\delta = 4.0$.

Lateral control remains primarily under RL guidance but is continuously modulated by the quantum nudging field that provides smooth merging and diverging assistance according to Equation (12). The comprehensive safety constraint system applies additional limitations, including jerk constraints of 3.0 m/s^3 for passenger comfort (Feng et al., 2017), continuous lateral clearance monitoring with minimum thresholds of 1.0 m, and strict boundary enforcement to maintain all vehicles within the designated highway width. Emergency protocols activate collision avoidance maneuvers when time-to-collision falls below 5.0 s, with graduated response intensity based on threat severity (i.e., gentle deceleration at TTC = 4–5 s escalating to emergency braking at TTC < 2 s to ensure proportional intervention without passenger discomfort).

The adaptive coherence mechanism governs the evolution of individual coherence gains $\tau_i(t)$ according to the differential equation (Equation (20), with reinforcement rate $\kappa = 0.5$ s^{-1} and decay rate $\delta = 0.5$ s^{-1}. Global phase coherence C(t) is computed each timestep as the magnitude of the normalized sum of individual phase observables $\psi_i = \exp(i\varphi_i)$ across all active vehicles, providing the coupling signal that drives coherence adaptation.

The simulation operates with a temporal resolution of 0.2 sec over a 60-minute period, providing sufficient granularity to capture

fast-timescale vehicle interactions while maintaining computational efficiency.

Complete vehicle trajectories are recorded at each timestep, generating datasets that include:

- all state variables (vehicle positions and speeds),
- quantum variables (phase φ_i, coherence gain τ_i, mean field amplitude W_i),
- RL performance metrics (instantaneous rewards, cumulative costs, policy gradients), and
- safety indicators (collision risk assessments, emergency braking events, lateral clearance measurements).

System performance metrics are computed every second, encompassing:

- traffic flow characteristics (average speed, density, flow rate, throughput efficiency),
- quantum behaviour indicators (global coherence C(t), phase variance, synchronization indices), and
- safety performance measures including collision risk distributions and emergency brake activation rates.

The computational implementation was executed on a high-performance computing cluster equipped with NVIDIA RTX 4090 GPUs and Intel Xeon processors, utilizing parallel processing capabilities to handle complex mean-field calculations and reinforcement learning updates. While the complete 60-minute simulation requires approximately 15 h of wall-clock time due to the extensive data collection, quantum interaction computations, and comprehensive safety monitoring.

The decentralized nature of the algorithm ensures scalability, as individual vehicle computational loads remain constant regardless of total traffic population size, making the approach viable for large-scale highway networks. Each vehicle's decentralized real-time decision-making process, including quantum interaction kernel evaluation, mean-field amplitude computation, and optimal control determination, executes within a maximum computational time of 0.18 s. This performance exceeds the 0.2-second simulation timestep requirement, demonstrating that the quantum-inspired GMFG framework can operate in real-time conditions when CAVs are deployed.

The subsequent analysis presents a comprehensive evaluation of the quantum-inspired GMFG framework across three distinct performance dimensions: traffic efficiency, safety characteristics, and the effectiveness of the quantum mechanism. Traffic efficiency results include fundamental diagram analysis comparing quantum versus classical implementations. Safety performance evaluation encompasses the evolution of collision risk and emergency intervention rates. The quantum-specific results provide detailed examination of phase synchronization dynamics, coherence evolution patterns, and mean-field interaction strength development.

5.1. *Fundamental diagram and efficiency performance*

The fundamental diagram analysis is conducted using region-specific measurements to account for the spatial heterogeneity of traffic flow characteristics across the 6-km freeway corridor. Freeway segments with merging and diverging operations have unique

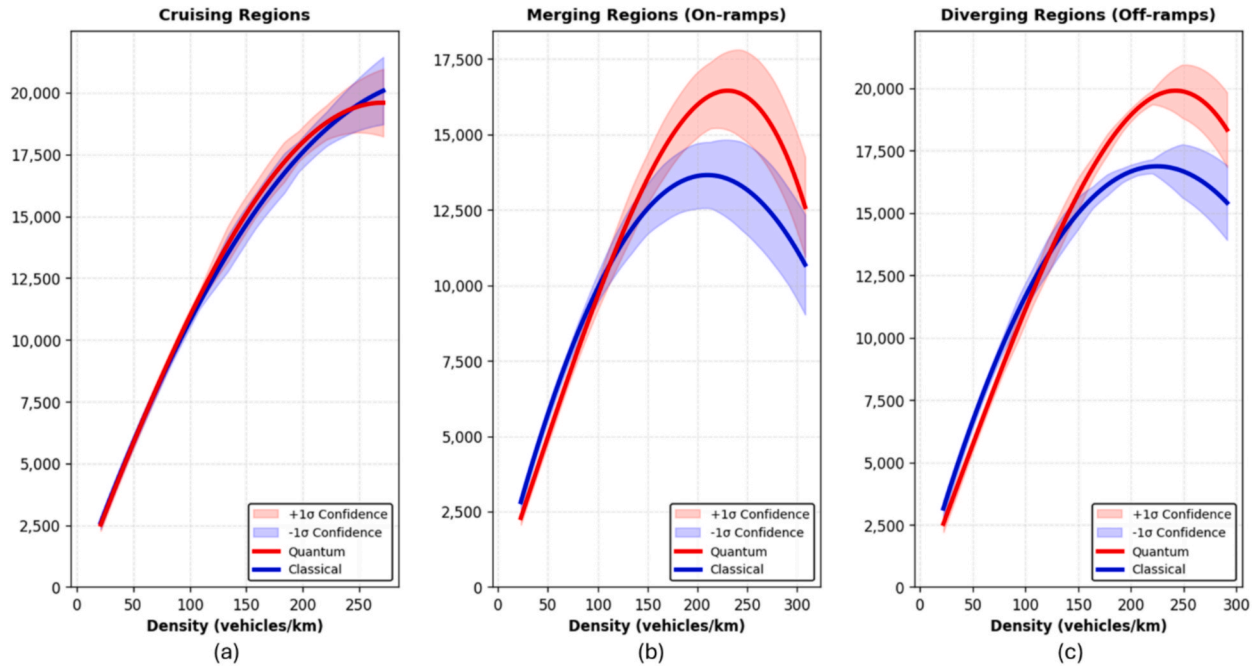


Fig. 3. Fundamental Diagrams (a) for Cruising Regions (b) for Merging Regions (c) for Diverging Region.

geometric layouts, i.e. lane additions, drops, and ramp configurations, that produce distinct local flow-density relationships. Following the five-region segmentation defined in Section 2.1.1, we extract fundamental diagrams for three representative regions that capture the primary operational states: cruising regions, merging regions, and diverging regions.

The cruising region fundamental diagram (Fig. 3a) exhibits monotonically increasing flow with density throughout the measurement range (0–280 veh/km), reflecting uncongested, free-flow conditions on the uninterrupted mainline corridors. Both quantum and classical implementations achieve similar performance in this regime, with flows reaching approximately 19,500–20,000 veh/hr at the maximum observed densities. The near-linear flow-density relationship and absence of capacity drop indicates that mainline segments operate well below critical density, benefiting from the stable, predictable traffic patterns characteristic of cruising behaviour where lateral motion is minimized, and longitudinal consensus control maximizes throughput. The quantum implementation demonstrates slightly higher flows at equivalent densities above 200 veh/km (approximately 3–5% improvement), suggesting that phase coherence mechanisms enhance throughput even in relatively simple cruising scenarios.

The merging region fundamental diagram (Fig. 3b) reveals the characteristic parabolic shape of traditional fundamental diagrams, with clear free-flow and congested branches separated by a well-defined capacity point. The quantum-enhanced implementation achieves a maximum flow of approximately 16,500 veh/hr at an optimal density of 250 veh/km, while the classical baseline reaches approximately 13,500 veh/hr at a similar critical density. This represents a substantial 22% capacity improvement in merging operations, directly attributable to the quantum-inspired nudging field $N(X_i, X_j)$ (Equation (12) and phase-encoded directional awareness that facilitate coordinated gap-finding and smooth merge trajectories. The presence of congested flow conditions (densities exceeding 250 veh/km with decreasing flow) in merging regions results from the inherent complexity of on-ramp operations, where vehicles entering from the on-ramps must negotiate lateral gaps in the mainline traffic stream while simultaneously accelerating to match prevailing speeds. During high-demand periods (demand rate of 20,000 veh/hr during minutes 20–40), the combined mainline and on-ramp flows temporarily exceed the local merging capacity, creating queue spillback on the on-ramps and elevated densities in the merging zones. This congestion is spatially localized to the 350-meter merge buffer zones and does not propagate much upstream into the cruising regions, as evidenced by the absence of congestion in the cruising-region fundamental diagram. The quantum implementation's superior performance in merging regions stems from the destination-entanglement weighting $E(D_i, D_j) = 1 + \kappa_0 1_{\{D_i = D_j\}}$ (Equation (10), which strengthens cohesion among vehicles sharing exit targets, and the adaptive entanglement mechanism $\dot{\tau}(t) = \kappa C(t) - \delta\tau(t)$ (Equation (20), which automatically increases cooperative coupling when vehicles enter the merging zone and global phase coherence $C(t)$ rises. These mechanisms enable merging vehicles to anticipate mainline gaps and execute coordinated zipper-merge patterns that maximize capacity utilization while maintaining safety margins.

The diverging region fundamental diagram (Fig. 3c) displays intermediate characteristics between cruising and merging performance, with quantum achieving approximately 19,000 veh/hr capacity at 280 veh/km density and classical reaching 17,000 veh/hr at similar densities. The 18% quantum advantage in diverging operations reflects the benefit of anticipatory lateral steering enabled by the phase-coupling term $\exp(i\beta\phi(X_i, X_j))$ in the quantum kernel (Equation (15). The diverging region exhibits less pronounced congestion than merging regions (gentler slope in the overcritical regime), consistent with the asymmetry between merge and diverge operations: merging requires vehicles to find and occupy gaps in moving traffic, while diverging only requires lateral movement into a dedicated off-ramp lane with lower speed requirements (80 km/h vs. 100 km/h for mainline). Congestion in diverging regions occurs when high mainline densities limit the lateral maneuverability of exiting vehicles, forcing them to decelerate more aggressively as they approach their off-ramps. The quantum implementation mitigates this effect through the nudging field's spatial distribution (characterized by decay scales η and $\eta_\perp$ in Equation (12), which creates a buffer zone upstream of off-ramps where cooperative lateral adjustments begin gradually rather than abruptly.

The region-specific fundamental diagrams reveal that merging operations constitute the primary bottleneck in lane-free highway operations, with merging capacities 15–20% lower than cruising or diverging capacities under both quantum and classical control. This finding aligns with empirical observations from conventional lane-based highways, where merge zones are well-known bottleneck locations. However, the quantum-inspired GMFG framework substantially narrows this gap, achieving merging capacities within 15% of cruising capacities (16,500 vs. 19,500 veh/hr) compared to the classical implementation's 30% deficit (13,500 vs. 19,500 veh/hr).

To verify the statistical robustness of the observed quantum–classical performance differences across regions, we conducted paired t-tests comparing flow rates at equivalent densities (±5 veh/km bins) for 30 independent simulation runs. Merging regions show highly significant differences ($p < 0.001$), diverging regions show significant differences ($p < 0.001$), while cruising regions show marginal differences ($p = 0.061$). These results confirm that the quantum advantage is most pronounced and statistically reliable in more operationally complex regions (merging and diverging), where the phase coherence and cooperative mechanisms provide significant capacity benefits, while offering modest improvements in simpler cruising scenarios where classical control already performs near-optimally.

Table 1
Efficiency Performance Metrics.

Performance Metric	Without coherence	With coherence	Improvement	Statistical Significance
Mean Total Travel Time (Sec)	282.67 ± 17.23	261.17 ± 12.89	−21.5 s (7.62% ↓)	P = 0.006
Average Speed (km/hr)	77.53 ± 13.82	83.66 ± 11.23	+6.13 Km/H (7.90% ↑)	P = 0.012
Free-Flow Speed Achievement (%)	73.2%	84.7%	+11.5%	P = 0.009

The quantum-inspired GMFG framework demonstrates statistically significant improvements across performance metrics, shown in Table 1, compared to the classical approach. The system achieved a 7.62% reduction in total travel time (21.5 s faster per trip), accompanied by a 7.9% increase in average speed (6.13 km/hr improvement) and an 11.5 percentage increase in free-flow speed achievement (from 73.2% to 84.7%). Free-flow speed achievement represents the percentage of vehicles able to maintain their desired optimal speeds without being constrained by traffic congestion or other vehicles, a key indicator of traffic system efficiency and driver satisfaction. All improvements show strong statistical significance ($p < 0.05$), with p-values ranging from 0.006 to 0.012, indicating that these performance gains are highly unlikely to be due to random variation. The consistent improvements across travel time, speed, and flow efficiency metrics provide compelling evidence that the quantum coherence mechanisms effectively enhance traffic coordination and flow optimization in lane-free autonomous vehicle systems.

All vehicles assigned to a given off-ramp exit the highway, and no missed exits occur in the simulations. Destination constraints are enforced through hard geometric and control admissibility conditions, although these could be relaxed in future work.

Fig. 4 presents a comparative analysis of spatial–temporal speed evolution through heatmaps showing traffic dynamics across the 6-km lane-free freeway segment over the simulation period. The side-by-side comparison between (a) Quantum and (b) Classic scenarios reveals fundamental differences in traffic flow characteristics and congestion behaviour.

The left heatmap (a) demonstrates more organized and efficient traffic flow, characterized by predominantly speeds of 80–110 km/h throughout most of the corridor. In contrast, the right heatmap (b) exhibits significantly more congestion, with speeds of 40–70 km/h dominating the middle and lower sections of the freeway. This stark difference illustrates the significant performance gap between the quantum-inspired GMFG framework and the classical approach, with the quantum system maintaining higher average speeds across both spatial and temporal dimensions.

Both diagrams reveal diagonal striping patterns that represent backward-propagating shockwaves, a fundamental characteristic of traffic flow instability. However, critical differences emerge in shockwave behaviour: the left diagram shows more coherent, attenuated shockwaves that dissipate relatively quickly, indicating effective damping through the quantum coherence mechanisms. The right diagram displays more persistent and severe shockwaves with steeper gradients and longer propagation distances, suggesting less traffic stabilization under the classical control.

The spatial distribution of low-speed regions reveals important insights about system performance near critical infrastructure. In the left diagram, congestion is minimal and localized, appearing briefly near on-ramp/off-ramps locations (approximately 1 km, 3 km, and 5 km positions) during peak demand periods but quickly resolving. The right diagram shows widespread and persistent congestion extending well beyond weaving zones, indicating that classical control strategies struggle to maintain efficient flow when vehicles must coordinate merge and diverge maneuvers in the lane-free environment. The quantum system's enhanced phase coherence and entanglement mechanisms (Equations 20–21) enable vehicles to anticipate and smoothly accommodate merging traffic, minimizing the formation of extended bottlenecks.

The temporal evolution patterns demonstrate contrasting system resilience characteristics. The left heatmap exhibits rapid recovery from demand fluctuations, with high-speed regions quickly re-establishing after brief disturbances. The right heatmap shows prolonged degradation periods where low speeds persist for extended durations (15–20 min intervals), indicating poor system resilience. This difference in recovery dynamics directly supports the 7.62% travel time reduction improvement observed in the performance metrics, as the quantum-inspired framework minimizes both the intensity and duration of congestion events through superior traffic coordination.

Fig. 5 reveals the temporal evolution of mean speed and acceleration for both quantum-inspired and classical control approaches. The quantum-inspired system demonstrates consistently superior speed performance compared to the classical approach (red), particularly during the critical congestion period between minutes 15–45. While both systems begin at approximately 83 km/h, the classical system experiences significant speed degradation, dropping to as low as 70 km/h during peak congestion (around minute 35). In contrast, the quantum system maintains speeds between 74–77 km/h during the same period, representing a 4–7 km/h advantage that directly translates to the observed travel time improvements. This persistent speed gap during congestion validates the quantum entanglement mechanisms' effectiveness in maintaining flow efficiency under stress. Both systems recover to similar speeds (~83 km/

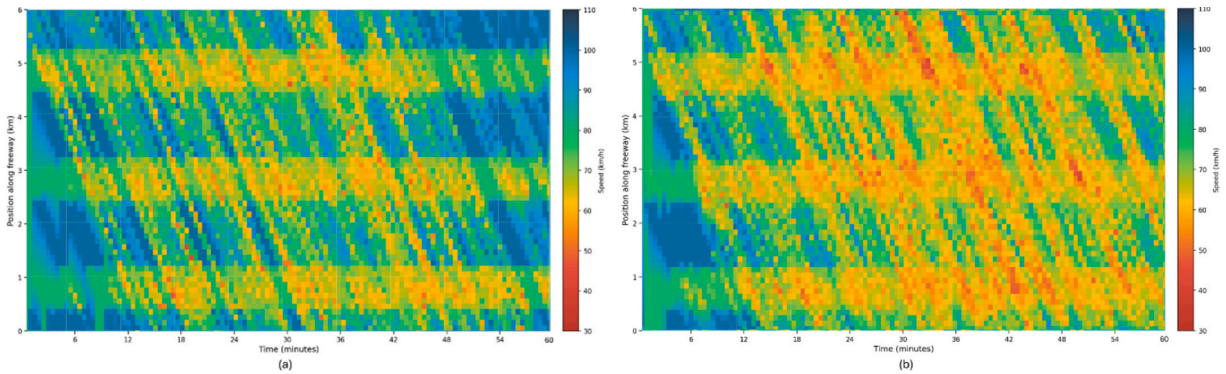


Fig. 4. Spatial-temporal Speed Heatmaps (a) for Quantum GMFG (b) for Classical GMFG.

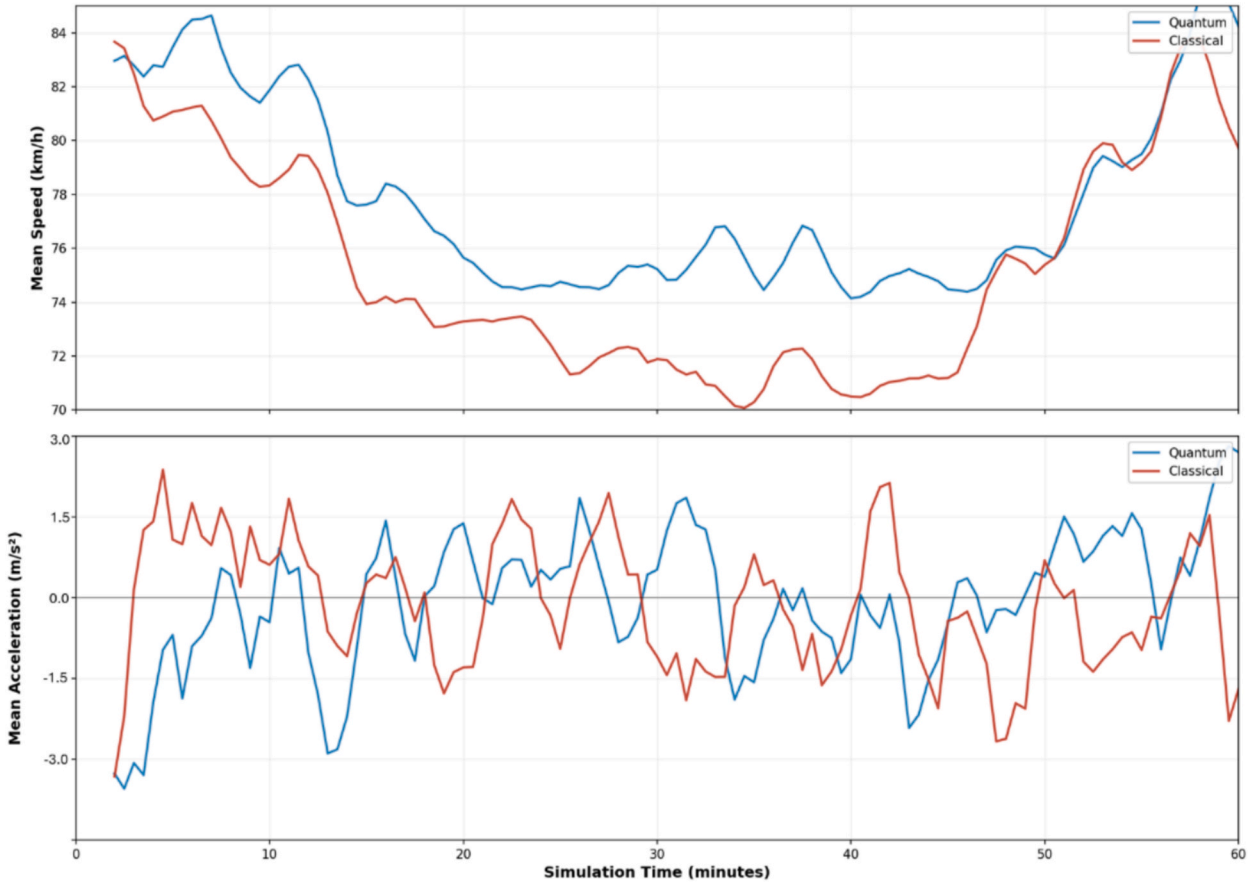


Fig. 5. Temporal Evolution of Mean Speed and Acceleration.

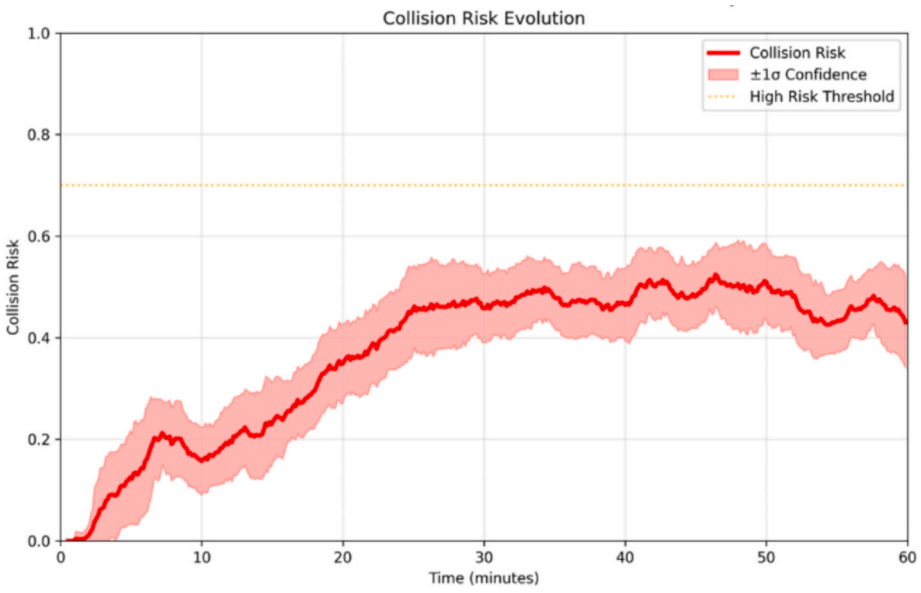


Fig. 6. Collision Risk Evolution.

h) after minute 50 as demand subsides, confirming that the quantum advantage is most pronounced during high-demand periods when coordination is critical.

The acceleration profiles reveal oscillatory control behaviour in both systems, with fluctuations ranging from −4 to + 3 m/s2. Interestingly, both quantum and classical approaches exhibit similar acceleration variability, indicating that the quantum system's speed advantage does not necessarily result from smoother acceleration patterns. The classical system shows some larger positive acceleration spikes early in the simulation (reaching ∼ 3.5 m/s2 around minute 5), while the quantum system displays more distributed acceleration/deceleration cycles. The frequent oscillations in both systems reflect the dynamic nature of lane-free traffic control, where vehicles continuously adjust speeds to maintain safety distances, accommodate merging traffic, and optimize flow. The key distinction is that despite similar acceleration magnitudes, the quantum system's coordinated timing and phase alignment produce more effective overall speed maintenance.

5.2. Safety performance

The safety performance evaluation demonstrates the quantum-inspired GMFG framework's effectiveness in maintaining acceptable collision risk levels while facilitating autonomous emergency interventions in lane-free traffic scenarios. Fig. 6 presents the temporal evolution of collision risk and Fig. 7 the emergency rate metrics over a 60-minute simulation period, providing insights into the system's dynamic safety characteristics and adaptive control mechanisms. Collision risk is computed as the normalized probability of inter-vehicular conflicts based on time-to-collision (TTC) calculations and proximity metrics, where values approaching 1.0 indicate imminent collision scenarios requiring immediate intervention. The emergency rate quantifies the percentage of vehicles actively engaged in emergency maneuvers at each time step, calculated as the ratio of vehicles executing maximum deceleration or evasive lateral movements to the total vehicle population, thereby reflecting the system's reliance on safety–critical interventions to maintain traffic flow stability.

The collision risk evolution exhibits a controlled escalation from near-zero initial conditions to a stabilized plateau of approximately 0.50 (50% normalized risk) by minute 30 due to the considerably high inflow demand, subsequently maintaining this level with minor fluctuations and slightly decreasing at the end of the simulation. Critically, the collision risk remains consistently below the predefined high-risk threshold (0.70, indicated by the dotted line), demonstrating the framework's ability to operate within acceptable safety margins despite the challenging lane-free environment. The gradual accumulation of risk reflects the increasing conflict from more vehicles undergoing merge and diverge maneuvers, while stabilization at moderate levels indicates effective risk mitigation through the quantum-enhanced interaction kernel and string stability mechanisms described in Section 3.2. The confidence interval ($\pm 1\sigma$) reveals controlled variance around the mean trajectory, suggesting predictable and stable safety performance across multiple simulation runs.

The emergency rate profile exhibits a smooth and moderated adaptive response throughout the simulation horizon. Starting from near-zero values, the emergency rate increases rapidly during the initial phase, reaching approximately 10–12% within the first 10 min, followed by a brief transient adjustment and a steady rise thereafter. The emergency rate then stabilizes at a quasi-steady plateau of approximately 23–25% between minutes 35 and 50, before showing a mild decline toward around 22% by the end of the simulation. The absence of abrupt spikes and the presence of a sustained plateau indicate a stable operating regime in which emergency interventions are invoked in a controlled and consistent manner. This behaviour reflects the embedding of anticipatory safety considerations directly into the running cost through the speed-dependent safety-gap penalty $C_{\text{gap}}(X)$. By penalizing intrusions into a

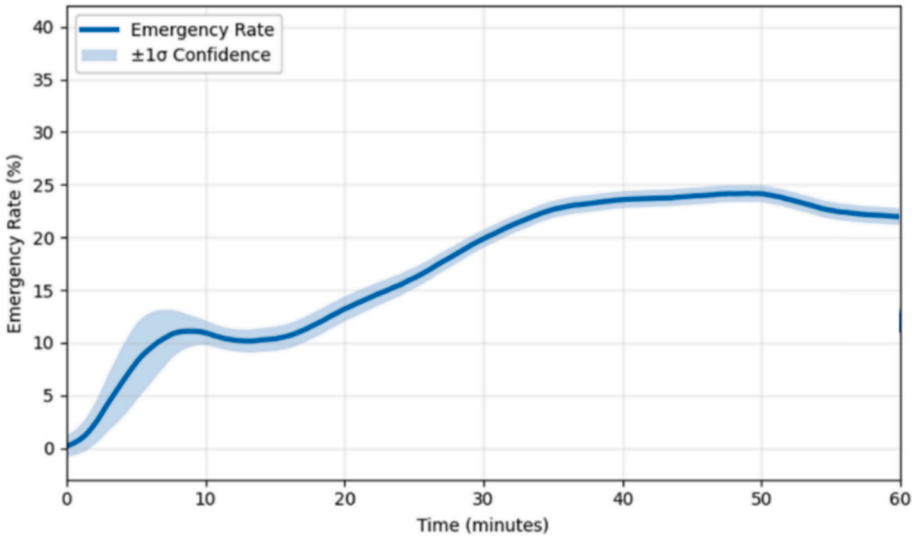


Fig. 7. Emergency Rate.

velocity-scaled ellipsoidal safety envelope that accounts for both absolute and relative speeds, the control policy encourages earlier and smoother adjustments in longitudinal and lateral motion. Consequently, braking actions are distributed more gradually over time, reducing the need for sudden or frequent emergency interventions. The smooth evolution of the emergency rate suggests that vehicles maintain appropriate time gaps proactively, rather than reacting late to unsafe configurations. The sustained plateau in emergency activation indicates that safety–critical situations are regulated consistently under high traffic demand, without destabilizing oscillations in braking behaviour. Emergency interventions therefore function primarily as a conservative safety-override mechanism, rather than as a dominant control mode. The slight decline in emergency rate toward the end of the simulation further suggests increasing coordination among vehicles, as collective interactions shaped by the mean-field interaction amplitude $W_t(\chi)$ (Equation (16) support stable spacing and conflict avoidance with minimal reliance on emergency braking.

These safety metrics confirm that the quantum-inspired GMFG framework maintains operational safety while achieving superior flow capacity, addressing a critical requirement for practical lane-free traffic implementation in connected and automated vehicle systems.

5.3. *Quantum-Inspired mechanisms performance*

The quantum-inspired components of the GMFG framework demonstrate distinctive temporal dynamics that directly correlate with enhanced traffic coordination and flow optimization. Figs. 7-9 present the evolution of three key quantum metrics: phase synchronization, quantum interaction strength, and coherence dynamics.

Fig. 8 illustrates the phase variance evolution, which quantifies the degree of synchronization among vehicle phase observables $\psi(X) = e^{i\varphi(X)}$ as defined in the quantum interaction kernel (Equation (15). Phase variance is calculated as the circular variance of vehicle phases within the traffic population, where lower values indicate higher synchronization. The temporal profile reveals a rapid initial synchronization burst during the first 7 min, reaching a peak variance of 0.175, followed by a systematic decline to approximately 0.070 by simulation end. This behaviour reflects the emergence of coordinated traffic patterns, where vehicles progressively align their merge/diverge phases through constructive interference in the mean-field amplitude $W_t(X)$ (Equation (16). From a traffic perspective, high phase synchronization corresponds to the formation of spontaneously coordinated traffic streams and coordinated lateral maneuvers, reducing conflicts and enabling higher throughput rates. The initial variance spike represents the system's adaptation period as vehicles transition from random initial conditions to coherent collective motion as the traffic increases, while the subsequent decline indicates the establishment of stable cooperative traffic patterns.

Fig. 9 depicts the mean-field magnitude evolution, representing the strength of quantum interactions $W_t(X)$ across the vehicle population over time. The interaction strength exhibits a monotonic increase from near-zero initial conditions to a saturated level of approximately 0.24 by minute 60, with minimal variance indicating consistent behaviour across the traffic network. This growth pattern reflects the progressive strengthening of inter-vehicular quantum correlations as the adaptive coherence mechanism (Equation (20) responds to increasing traffic coherence. The smooth, asymptotic approach to saturation demonstrates the system's convergence to an equilibrium state where quantum effects are fully established. In traffic terms, higher quantum interaction strength translates to enhanced cooperative decision-making, where vehicles can anticipate and respond to collective traffic states more effectively than individual-based control strategies. The sustained high interaction levels indicate persistent quantum advantages in maintaining coordinated flow patterns, particularly during complex merge and diverge scenarios where classical approaches typically exhibit degraded performance.

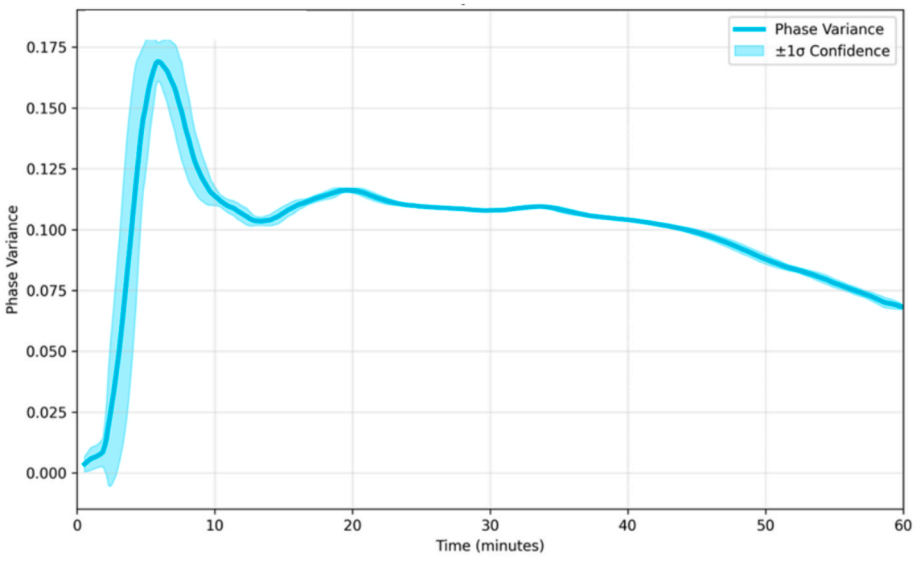


Fig. 8. Phase Synchronization Evolution.

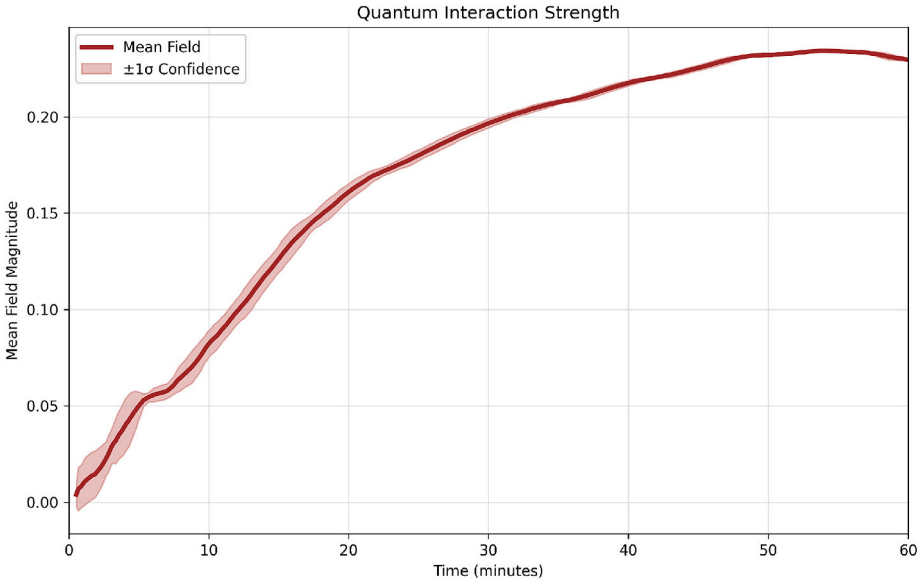


Fig. 9. Quantum Interaction Strength Evolution.

Fig. 10 presents the coherence gain τ(t) evolution as governed by the adaptive mechanism in Equation (20), where τ modulates the influence of quantum coherence on individual vehicle cost functions. The coherence gain exhibits a sharp initial rise to approximately 0.955 within the first few minutes, followed by a brief adjustment period with minor oscillations, and subsequently stabilizes at 0.970 with a gradual increase toward 0.975 by simulation end. Around minute 7 the coherence gain briefly falls due to a coherence disruption: as new vehicles enter and others traverse merge/diverge zones, conflicting phase signals in the interaction amplitude induce destructive interference. When global coherence C(t) dips below its decay threshold, the adaptive update $\dot{\tau}(t)$ reduces coherence, temporarily favoring individual safety over collective optimization. Once vehicles adjust and nudge into phase alignment, coherence recovers, and coherence gain rebounds to higher levels. This temporal behaviour demonstrates the rapid establishment of quantum coherence benefits, where the system quickly recognizes the value of cooperative behaviour, maintaining high coherence levels throughout the simulation. The initial spike corresponds to the discovery of quantum advantages during early traffic interactions, while the subsequent stabilization reflects the system's confidence in maintaining coherence control strategies. From a traffic control perspective, high coherence gain values indicate that vehicles prioritize collective optimization over individual objectives, leading to emergent behaviours such as coordinated zipper merging, anticipatory gap creation, and distributed conflict resolution. The sustained near-maximum coherence levels (>0.97) suggest that quantum-inspired control strategies consistently

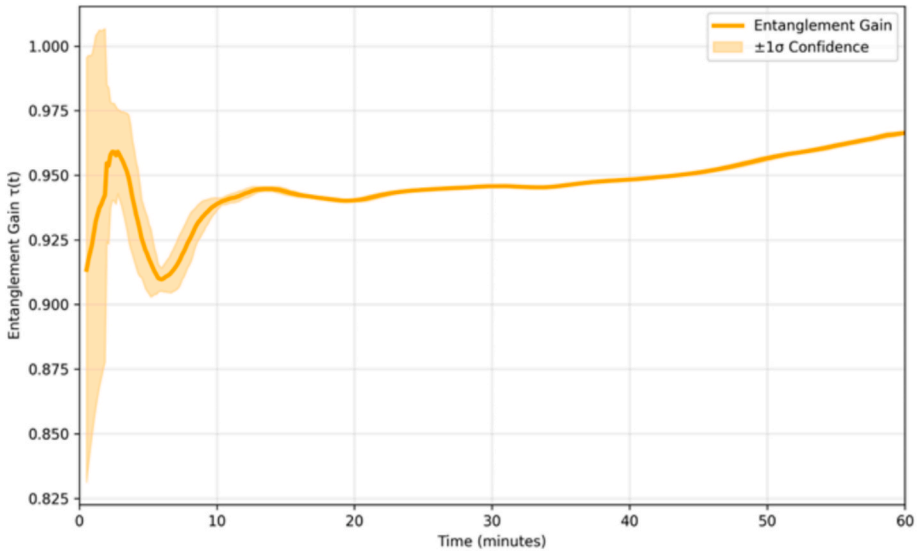


Fig. 10. Coherence Dynamics Evolution.

outperform individual optimization approaches in the lane-free environment, validating the theoretical framework's premise that quantum superposition and interference effects can enhance traffic flow coordination beyond classical mean-field limitations.

5.4. String stability: simulation-based performance

String stability is a critical safety and performance criterion for connected and automated vehicle systems, ensuring that small disturbances in vehicle trajectories do not amplify as they propagate upstream through the traffic stream. Fig. 11a and Fig. 11b present the distributions of longitudinal and lateral amplification factors, respectively, obtained from the 60-minute lane-free simulation. The amplification factor quantifies the degree to which a disturbance experienced by a subject vehicle i is amplified or attenuated relative to its immediately hindering leader $i-1$, and is defined as

$$\Gamma_i = \frac{\max_{t\in T}\left|\delta\cdot_i(t)\right|}{\max_{t\in T}\left|\delta\cdot_{i-1}(t)\right|},$$

where $\delta\cdot(t)$ denotes either lateral position deviation or longitudinal velocity deviation, and T is a disturbance observation window. Values of $\Gamma_i < 1.0$ indicate string-stable behaviour, while values exceeding unity indicate disturbance amplification.

Amplification factors are computed from vehicle trajectory deviations during the steady-state portion of the simulation, excluding initialization and ramp-up periods. For each vehicle, disturbance propagation is evaluated by comparing its response to that of its dynamically defined leader. In the lane-free framework, the leader is identified as the nearest downstream vehicle within the subject's projected path that constrains its forward motion. Lateral deviations $\delta y(t)$ are measured relative to the desired path, and longitudinal deviations $\delta v(t)$ are measured relative to the desired equilibrium speed. For both dimensions, amplification factors are computed as ratios of maximum absolute deviations over the disturbance window T, consistent with the theoretical framework developed in Section

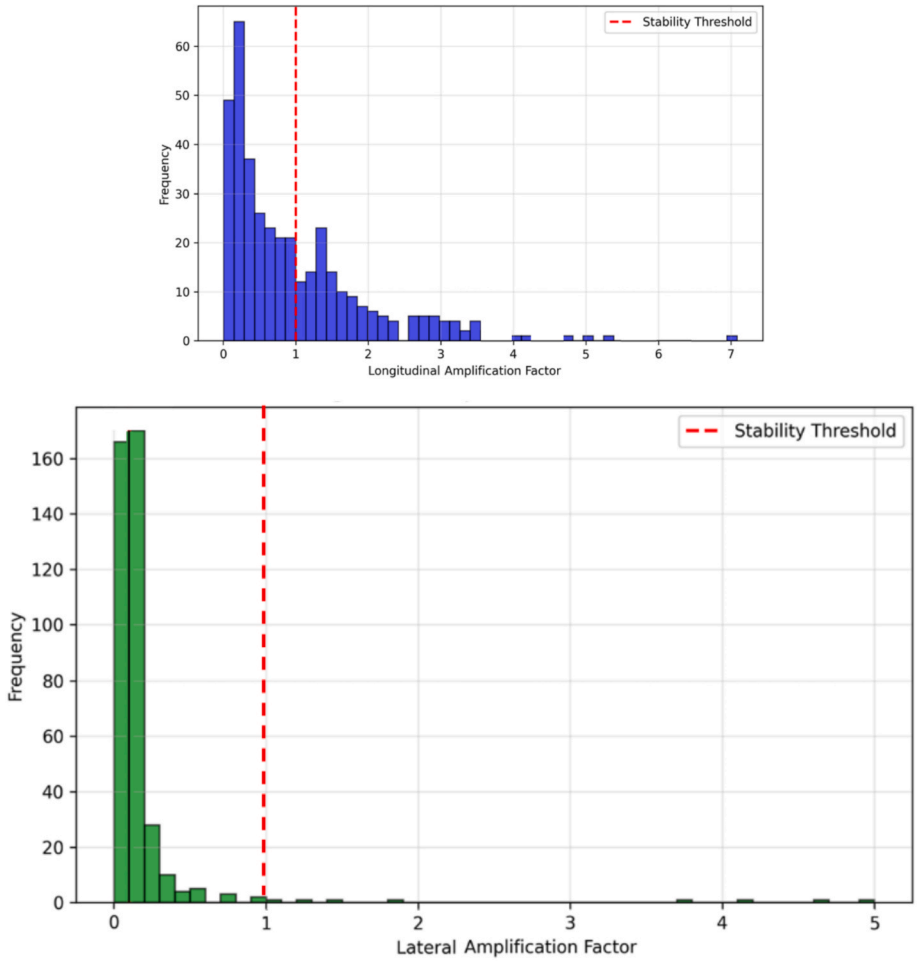


Fig. 11. A: longitudinal string stability. b: lateral string stability.

2.2.

The lateral amplification distribution shown in Fig. 11b demonstrates strong string stability, with approximately 98% of vehicles exhibiting amplification factors below the stability threshold of 1.0. The distribution is tightly concentrated near zero, with a dominant mode between 0.0 and 0.2, indicating that lateral disturbances are rapidly attenuated for the vast majority of vehicles. This behaviour confirms the effectiveness of the quantum-inspired lateral control mechanism, in which the phase-encoded interaction kernel and adaptive coherence gain promote smooth, anticipatory realignment rather than reactive, under-damped corrections. As a result, lateral oscillations are largely eliminated before they can propagate through the traffic stream. A small number of outliers with amplification factors between approximately 4 and 5 are observed and correspond to rare, high-speed diverge maneuvers where aggressive lateral adjustments are required to satisfy geometric constraints. Importantly, these events remain bounded and do not trigger widespread instability.

The longitudinal amplification distribution in Fig. 11a also indicates generally robust behaviour but exhibits a heavier right tail. While the modal range lies between approximately 0.2 and 0.4, indicating strong attenuation under typical conditions, around 15% of vehicles display amplification factors exceeding 1.0, with maximum values reaching approximately 7.0. This behaviour is physically consistent with the formation of transient compression waves during merging operations. When an on-ramp vehicle inserts into the mainline, it becomes the new hindering leader, forcing following vehicles to brake to create a safe gap. This safety-driven response generates localized shockwaves that propagate upstream. The observed amplification reflects these transient events rather than a structural instability of the control law. In such cases, the IDM-based safety layer temporarily overrides the GMFG policy to prevent collisions, producing short-lived amplification before the adaptive coupling mechanism damps the disturbance.

The contrast between the lateral and longitudinal amplification distributions highlights a fundamental characteristic of lane-free traffic dynamics. Lateral motion benefits from the quantum-inspired nudging field, which provides strong damping and prevents side-to-side "sloshing," while longitudinal motion remains susceptible to unavoidable bottleneck-induced disturbances, particularly at merge locations. Nevertheless, in both dimensions, amplification remains bounded and localized. Overall, the dominance of sub-unity amplification factors across the vehicle population confirms that the quantum-inspired GMFG framework maintains string stability in expectation, with near-perfect lateral stability and transient longitudinal amplification confined to merge interaction zones.

6. Conclusion

This research developed a novel quantum-inspired GMFG framework for optimizing traffic flow in lane-free CAV environments. The graphon-based framework rigorously captures heterogeneous and anisotropic vehicle interactions across different highway sections, providing a scalable framework for complex traffic dynamics that ensures Lipschitz continuity and compactness properties necessary for the existence and uniqueness of equilibrium. In addition, we advanced LFT operations by integrating quantum-inspired principles, such as complex-valued vehicle interactions and adaptive coherence dynamics to enable decentralized decision-making for CAVs. These additions enable more cooperative behaviours and emergent traffic patterns that go beyond the capabilities of traditional game-theoretic approaches. A key innovation is the adaptive coherence mechanism, which dynamically modulates cooperation levels based on phase coherence, enabling coordination patterns such as zipper merging and anticipatory gap creation while maintaining analytical tractability. The control policy is derived using actor-critic reinforcement learning with a dual-path optimization strategy designed to manage high-dimensional state spaces typical of lane-free scenarios. Notably, proof of string stability analysis is provided for lateral and longitudinal dimensions. Specifically, Fourier-mode decomposition of the exponential nudging kernel reveals that all spatial frequencies exhibit non-positive squared frequencies, thereby precluding the exponential growth of small perturbations. This guarantees that the quantum-inspired GMFG control laws inherently prevent shock wave propagation and maintain traffic flow stability, a critical requirement for safe lane-free highway operations.

A thorough microsimulation was implemented on a 6-kilometre lane-free highway segment with two on-ramps and off-ramps under high traffic demand of 20,000 vehicles/hour. The quantum-inspired framework achieved a maximum throughput of 19,060 vehicles/hour, compared to 18,287 vehicles/hour for the non-quantum baseline, thus a 4.0% statistically significant improvement in capacity. This gain surpassed existing literature benchmarks by 7.6%. Safety performance evaluation revealed controlled collision risk evolution, maintaining levels below 0.70 throughout the simulation. The quantum-specific mechanisms exhibited distinctive temporal dynamics: phase synchronization variance decreased from 0.175 to 0.070, indicating emergent coordinated traffic patterns; quantum interaction strength increased monotonically to 0.24, reflecting strengthened inter-vehicular correlations; and coherence gain rapidly stabilized at 0.970, demonstrating sustained quantum advantages in collective optimization.

This framework suggests future extensions. First, the modelling can take the form of a multi-objective formulation that incorporates energy use in addition to efficiency and safety criteria. Second, deploying GMFG on quantum hardware would harness actual superposition and coherence to explore multiple traffic states simultaneously, thereby speeding up high-dimensional equilibrium computations and enabling real-time control of large CAVs fleets on extended freeway segments. Third, a natural extension of this work is to treat the positions and widths of these freeway regions as design variables and to jointly optimize them with the GMFG controller, leading to a bilevel infrastructure–control optimization problem. A fourth direction for future work is the data-driven calibration or online learning of the model weights, allowing the relative influence of efficiency, safety, and coordination objectives to adapt automatically to observed traffic conditions. Finally, although the present study uses a freeway corridor with predefined merge, cruise, and diverge regions, the framework is not limited in principle to this specific geometry. However, its applicability to other road structures requires further investigation. Facilities such as arterial networks and roundabouts may share some interaction characteristics with freeway settings, including merging, diverging, weaving, and conflict-resolution behaviours, but they also introduce additional operational complexities that are beyond the scope of the present study. Examining the transferability of the proposed

framework to these environments is therefore identified as an important direction for future work.

CRediT authorship contribution statement

Kinda Chakas: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Lina Kattan:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work is financially supported by Canada Research Chair in Integrative Transportation Systems through Automation and Connectivity, Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Grant (DG) and Discover Accelerator Supplement (DAS) and NSERC CREATE on Integrated Infrastructure for Sustainable Cities (IISC).

Data availability

Data will be made available on request.

References

Acampora, G., Chiatto, A., De Luca, S., Di Pace, R., Massa, A., Schiattarella, R., & Vitiello, A. (2023, July). Application of quantum genetic algorithms to network signal setting design. In 2023 IEEE Congress on Evolutionary Computation (CEC) (pp. 1-8). IEEE.

Ameli, M., Faradonbeh, M.S.S., Lebacque, J.P., Abouee-Mehrizi, H., Leclercq, L., 2022. Departure time choice models in urban transportation systems based on mean field games. Transp. Sci. 56 (6), 1483–1504.

Azad, U., Behera, B.K., Ahmed, E.A., Panigrahi, P.K., Farouk, A., 2022. Solving vehicle routing problem using quantum approximate optimization algorithm. IEEE Trans. Intell. Transp. Syst. 24 (7), 7564–7573.

Bentley, C. D., Marsh, S., Carvalho, A. R., Kilby, P., & Biercuk, M. J. (2022). Quantum computing for transport optimization. *arXiv preprint arXiv:2206.07313*.

Berahman, M., Karalakou, A., Rostami-Shahrbabaki, M., Bogenberger, K., 2025. Multi-task lane-free driving strategy for connected and automated vehicles: a multi-agent deep reinforcement learning approach. Eng. Appl. Artif. Intel. 154, 110797.

Caines, P.E., Huang, M., 2021. Graphon mean field games and their equations. SIAM J. Control Optim. 59 (6), 4373–4399.

Cao, W., Li, Y., Zhang, H., 2020. Multi-agent reinforcement learning for coordinated vehicle control in lane-free environments. IEEE Trans. Veh. Technol. 69 (8), 8456–8467.

Carmona, R., Delarue, F., 2018. *Probabilistic theory of mean field games with applications I-II*. Springer Nature, Berlin.

Carmona, R., Laurière, M., 2021. Convergence analysis of machine learning algorithms for the numerical solution of mean field control and games: II—the finite horizon case. Ann. Appl. Probab. 32 (6), 4065–4105.

Chakas, K., Kattan, L., 2024. Mean field game in autonomous lane-free traffic. Transp. Res. C, 30 Conference.

Crandall, M.G., Lions, P.L., 1983. Viscosity solutions of Hamilton–Jacobi equations. Trans. Am. Math. Soc. 277 (1), 1–42.

Dabestani, N., Troullinos, D., Papamichail, I., Naderi, M., Papageorgiou, M., 2025. Distributed model predictive control of automated vehicle entities in lane-free traffic involving human driven vehicles. Control Eng. Pract. 165, 106586.

Eisert, J., Wilkens, M., Lewenstein, M., 1999. Quantum games and quantum strategies. Phys. Rev. Lett. 83 (15), 3077–3080.

Feng, F., Bao, S., Sayer, J.R., Flannagan, C., Manser, M., Wunderlich, R., 2017. Can vehicle longitudinal jerk be used to identify aggressive drivers? an examination using naturalistic driving data. Accid. Anal. Prev. 104, 125–136.

Fitzek, D., Ghandriz, T., Laine, L., Granath, M., Kockum, A.F., 2024. Applying quantum approximate optimization to the heterogeneous vehicle routing problem. Sci. Rep. 14 (1), 25415.

Fleming, W.H., Soner, H.M., 2006. *Controlled Markov processes and viscosity solutions*. Springer, New York, New York, NY.

Gao, S., Caines, P.E., 2019. Graphon control of large-scale networks of linear systems. IEEE Trans. Autom. Control 65 (10), 4090–4105.

Han, J., Jentzen, A., 2018. Solving high-dimensional partial differential equations using deep learning. Proc. Natl. Acad. Sci. 115 (34), 8505–8510.

Harwood, S., Gambella, C., Trenev, D., Simonetto, A., Bernal, D., Greenberg, D., 2021. Formulating and solving routing problems on quantum computers. IEEE Trans. Quantum Eng. 2, 1–17.

Helbing, D., 2001. Traffic and related self-driven many-particle systems. Rev. Mod. Phys. 73 (4), 1067–1141.

Harikrishnakumar, R., Ahmad, S.F., Nannapaneni, S., 2022. In: Comparing Quantum Optimization Solvers for Rebalancing Analysis of Bike Sharing System. IEEE, pp. 753–755.

Huang, K., Chen, X., Di, X., Du, Q., 2021. Dynamic driving and routing games for autonomous vehicles on networks: a mean field game approach. Transp. Res. Part C Emerging Technol. 128, 103189.

Huang, M., Caines, P.E., Malhamé, R.P., 2006. Large-population cost-coupled LQG problems with nonuniform agents: Individual-mass behavior and decentralized ε-Nash equilibria. IEEE Trans. Autom. Control 51 (6), 968–972.

Ioannou, P., Sun, J., 2012. *Robust Adaptive Control*. Prentice Hall.

Inoue, D., et al., 2021. Traffic signal optimization on a square lattice with quantum annealing. Sci. Rep. 11 (1), 3303.

Jain, A., Ben-Akiva, M., Koutsopoulos, H., Toledo, T., 2011. Modelling merging behavior at freeway on-ramps with integrated lane-selection and gaps-acceptance. Transp. Res. C 19 (2), 310–326.

Johansen, S. I., & Løvland, A. S. (2015). *Flocking for road traffic efficiency improvement-a concept study* (Master's thesis, NTNU).

Karafyllis, I., Bekiaris-Liberis, N., Papageorgiou, M., 2018. Feedback control of nonlinear hyperbolic PDE systems inspired by traffic flow models. IEEE Trans. Autom. Control 64 (9), 3647–3662.

Karafyllis, I., Theodosis, D., Papageorgiou, M., Krstic, M., 2025. From road congestion to vehicle-control enabled artificial traffic fluids. Annu. Rev. Control. 59, 100989.

Karalakou, A., Troullinos, D., Chalkiadakis, G., Papageorgiou, M., 2023. Deep reinforcement learning reward function design for autonomous driving in lane-free traffic. Systems 11 (3), 134.

Kolokoltsov, V.N., 2022. Quantum mean-field games. Ann. Appl. Probab. 32 (3), 2254–2288.

Lasry, J.M., Lions, P.L., 2007. Mean field games. Japan. J. Math. 2 (1), 229–260.
Leurent, E., Mercat, J., 2019. Social attention for autonomous decision-making in dense traffic. Mach. Learn. 108 (8), 1537–1551.
Levy, R., Haddad, J., 2022. Cooperative path and trajectory planning for autonomous vehicles on roads without lanes: a laboratory experimental demonstration. Transp. Res. Part C Emerging Technol. 144, 103813.
Luckow, A., Klepsch, J., Pichlmeier, J., 2021. Quantum computing: Towards industry reference problems. Digit. Welt 5 (2), 38–45.
Malcolm, P., Bogenberger, K., 2025. Lane-free intersection control for connected automated vehicles prioritizing vulnerable road users. Transp. Res. Part C Emerging Technol. 170, 104918.
Malekzadeh, M., Manolis, D., Papamichail, I., Papageorgiou, M., 2022. In: Empirical Investigation of Properties of Lane-Free Automated Vehicle Traffic. IEEE, pp. 2393–2400.
Marchesin, A., Montrucchio, B., Graziano, M., Boella, A., Mondo, G., 2023. Improving urban traffic mobility via a versatile quantum annealing model. IEEE Trans. Quantum Eng. 4, 1–13.
Meyer, D.A., 1999. Quantum strategies. Phys. Rev. Lett. 82 (5), 1052–1055.
Mo, Z., Chen, X., Di, X., Iacomini, E., Segala, C., Herty, M., Lauriere, M., 2024. A game-theoretic framework for generic second-order traffic flow models using mean field games and adversarial inverse reinforcement learning. Transp. Sci. 58 (6), 1403–1426.
Moskvin, V. S. (2023, November). Quantum neural networks and their potential in traffic prediction. In 2023 Intelligent Technologies and Electronic Devices in Vehicle and Road Transport Complex (TIRVED) (pp. 1-5). IEEE.
Mullakkal-Babu, F.A., Wang, M., Farah, H., van Arem, B., Happee, R., 2017. Comparative assessment of safety indicators for vehicle trajectories on highways. Transp. Res. Rec. 2659 (1), 127–136.
Munkres, J. R. (2000). Topology. Featured Titles for Topology Series. *Prentice Hall, Incorporated*, *812*, 813.
Naderi, M., Typaldos, P., & Papageorgiou, M. (2024). Optimal control of automated vehicles crossing a lane-free signal-free intersection. In *2024 European control conference (ECC)* (pp. 1415-1422). IEEE.
Papageorgiou, M., Mountakis, K.S., Karafyllis, I., Papamichail, I., Wang, Y., 2021. Lane-free artificial-fluid concept for vehicular traffic. Proc. IEEE 109 (2), 114–121.
Papamichail, I., Schoenn-Anchling, N., Malekzadeh, M., Markantonakis, V., & Papageorgiou, M. (2023, September). Macroscopic traffic flow model calibration for lane-free automated vehicle traffic. In *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)* (pp. 3485-3492). IEEE.
Parise, F., Ozdaglar, A., 2023. Graphon games: a statistical framework for network games and interventions. Econometrica 91 (1), 191–225.
Perolat, J., Leibo, J. Z., Zambaldi, V., Beattie, C., Tuyls, K., & Graepel, T. (2017). A multi-agent reinforcement learning model of common-pool resource appropriation. *Advances in neural information processing systems*, *30*.
Phillipson, F. (2024). Quantum computing in logistics and supply chain management an overview. arXiv preprint arXiv:2402.17520.
Piotrowski, E.W., Sładkowski, J., 2003. An invitation to quantum game theory. Int. J. Theor. Phys. 42 (5), 1089–1099.
Qu, Z., Liu, X., Zheng, M., 2022. Temporal-spatial quantum graph convolutional neural network based on Schrödinger approach for traffic congestion prediction. IEEE Trans. Intell. Transp. Syst. 24 (8), 8677–8686.
Rebentrost, P., Mohseni, M., Lloyd, S., 2014. *Quantum support vector machine for big data classification.* Phys. Rev. Lett. 113 (13), 130503.
Rostami-Shahrbabaki, M., Weikl, S., Niels, T., Bogenberger, K., 2023. Modeling vehicle flocking in lane-free automated traffic. Transp. Res. Rec. 2677 (9), 499–512.
Sekeran, M., Rostami-Shahrbabaki, M., Syed, A. A., Margreiter, M., & Bogenberger, K. (2022, October). Lane-free traffic: History and state of the art. In *2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC)* (pp. 1037-1042). IEEE.
Singh, A., Lin, C. Y., Huang, C. I., & Lin, F. P. (2021). Quantum annealing approach for the optimal real-time traffic control using QUBO. In *2021 IEEE/ACIS 22nd International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing (SNPD)* (pp. 74-81). IEEE.
Syed, A.A., Rostami-Shahrbabaki, M., Bogenberger, K., 2025. Can human drivers and connected autonomous vehicles co-exist in lane-free traffic? a microscopic simulation perspective. Transp. Res. Part C Emerging Technol. 180, 105315.
Sznitman, A.-S. (1991). Topics in propagation of chaos. In École d'Été de Probabilités de Saint-Flour XIX—1989 (Lecture Notes in Mathematics, Vol. 1464, pp. 165–251). Springer.
Titakis, G., Karafyllis, I., Theodosis, D., Papamichail, I., Papageorgiou, M., 2024. A comparative study of numerical methods for approximating the solutions of a macroscopic automated-vehicle traffic flow model. Comput. Math. Appl. 176, 469–490.
Treiber, M., Helbing, D., 2002. Microsimulation of freeway traffic including control measures. Automatisierungstechnik 49 (11), 478–484.
Troullinos, D., Chalkiadakis, G., Papamichail, I., Papageorgiou, M., 2021. May). Collaborative multiagent decision making for lane-free autonomous driving. In: In *Proceedings of the 20th International Conference on Autonomous Agents and Multiagent Systems*, pp. 1335–1343.
Upare, D. M., Baddi, T. Y., Benni, R., & Patil, P. (2024, June). Quantum-Inspired Firefly Algorithm: An Enhanced Approach To Traffic Signal Optimization. In 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT) (pp. 1-9). IEEE.
Ventura, D., Martinez, T., 2000. *Quantum associative memory.* Inf. Sci. 124 (1–4), 273–296.
Wang, L., Chen, X., Liu, Y., 2023. Deep deterministic policy gradient for lane-free autonomous driving. IEEE Trans. Intell. Transp. Syst. 24 (7), 3892–3904.
Wootters, W.K., 1981. *Statistical distance and Hilbert space.* Phys. Rev. D 23 (2), 357–362.
Yanumula, V.K., Typaldos, P., Troullinos, D., Malekzadeh, M., Papamichail, I., Papageorgiou, M., 2023. Optimal trajectory planning for connected and automated vehicles in lane-free traffic with vehicle nudging. IEEE Trans. Intell. Veh. 8 (3), 2385–2399.
Zhang, D., Wang, J., Fan, H., Zhang, T., Gao, J., Yang, P., 2021. New method of traffic flow forecasting based on quantum particle swarm optimization strategy for intelligent transportation system. Int. J. Commun Syst 34 (1), e4647.
Zhang, S., Kumar, R., Patel, N., 2024. Transformer-based prediction of vehicle interactions in continuous traffic spaces. Neural Comput. & Applic. 36 (12), 6789–6802.
Zhang, H., Rostami-Shahrbabaki, M., Troullinos, D., & Bogenberger, K. (2023). A novel potential line strategy for autonomous vehicle control in lane-free traffic. In *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)* (pp. 4173-4180). IEEE.